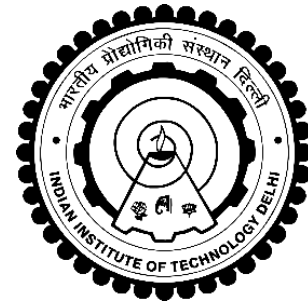




ELL 100 – Introduction to Electrical Engineering

LECTURE 3: ANALYSIS OF DC CIRCUITS

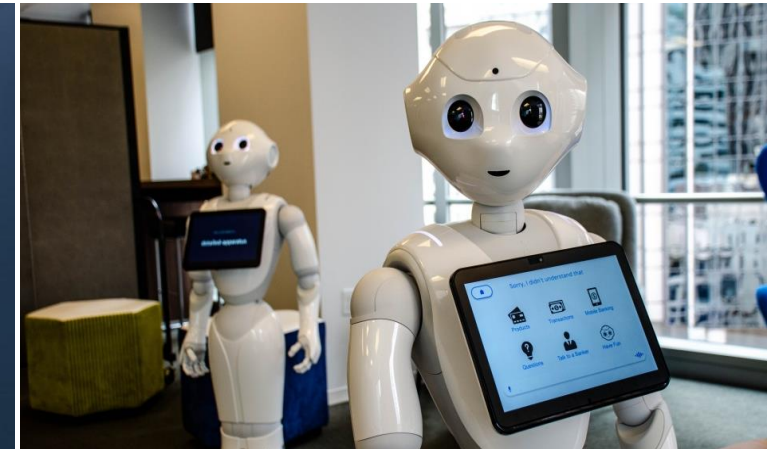
VOLTAGE/CURRENT SOURCES & SOURCE TRANSFORMATIONS

Outline

- ☐ Introduction
- ☐ Voltage and Current Sources
- ☐ Passive Sign Convention
- ☐ Series and Parallel Connection of Sources
- ☐ Node, Branches and Loops
- ☐ Series and Parallel Elements
- ☐ Source Transformation
- ☐ Exercise/Numerical Analysis
- ☐ Unsolved Problems
- ☐ References

INTRODUCTION

Electrical Circuits seem to be everywhere!

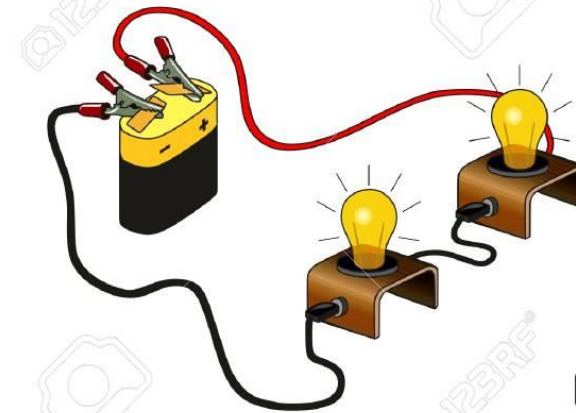


INTRODUCTION

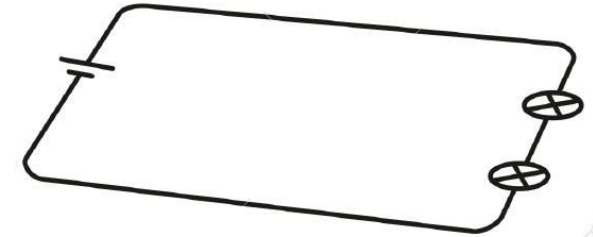
Analysis of Circuits implies:

- Finding voltage across any element of the circuit.
- Finding current through any element of the circuit.

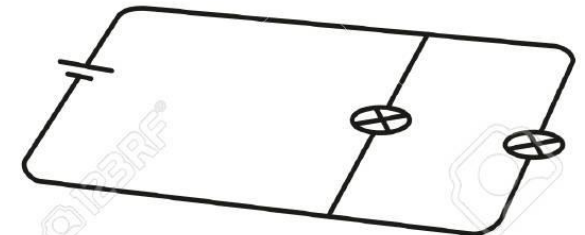
Electrical Circuit Types



Series Circuit



Parallel Circuit

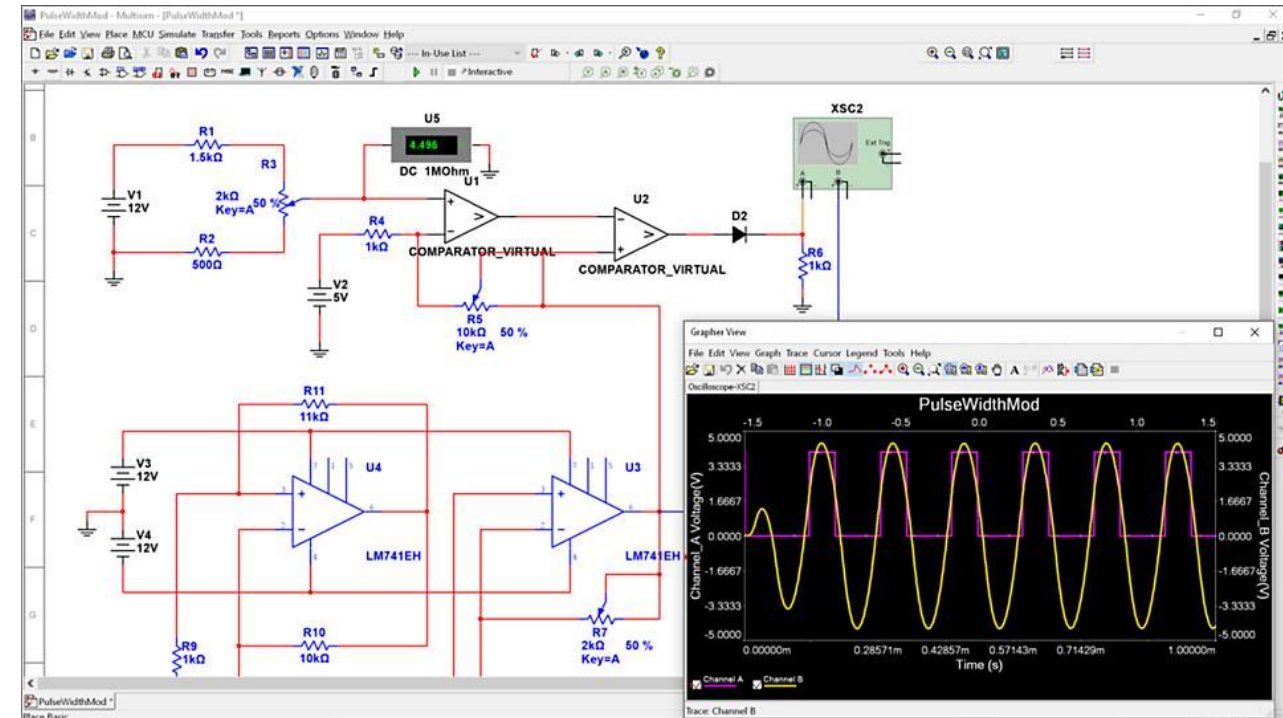


INTRODUCTION

Analysis of Circuits



Modern trains are powered by electric motors. Their electrical systems are best analyzed using AC or phasor analysis techniques.

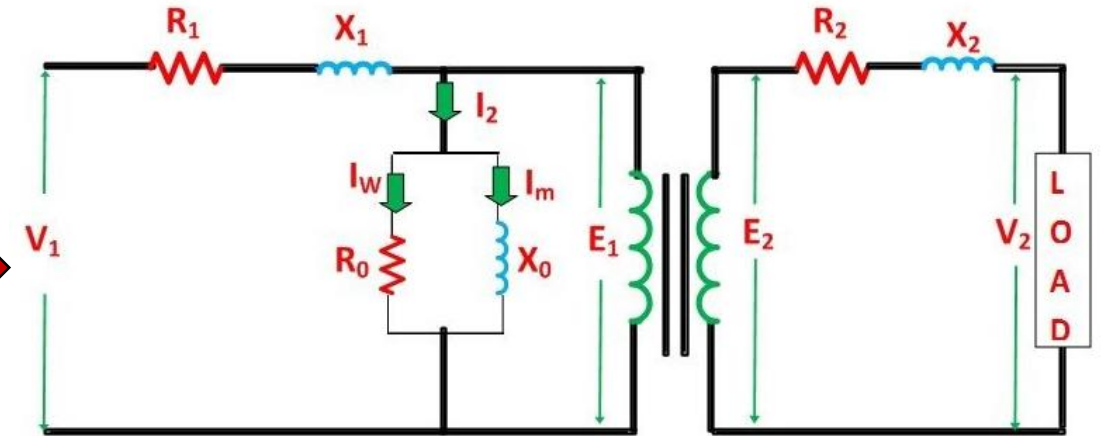


Knowledge of circuit analysis enable oneself to use circuit simulator developed for students to learn analog, digital, and power electronics.

INTRODUCTION



**Circuit representation of
a Physical Transformer**

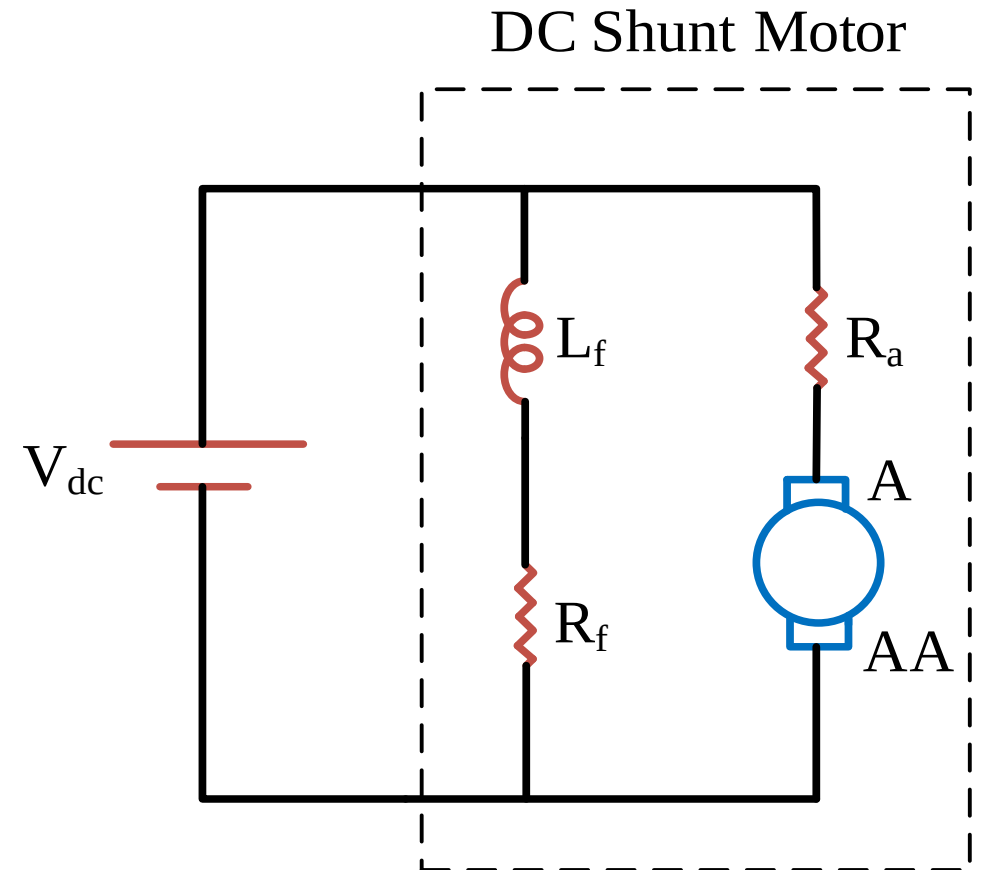


Knowledge of circuit analysis helps to analyse any electrical device with their equivalent circuit representation.

INTRODUCTION



**Circuit representation of
a DC Shunt Motor**

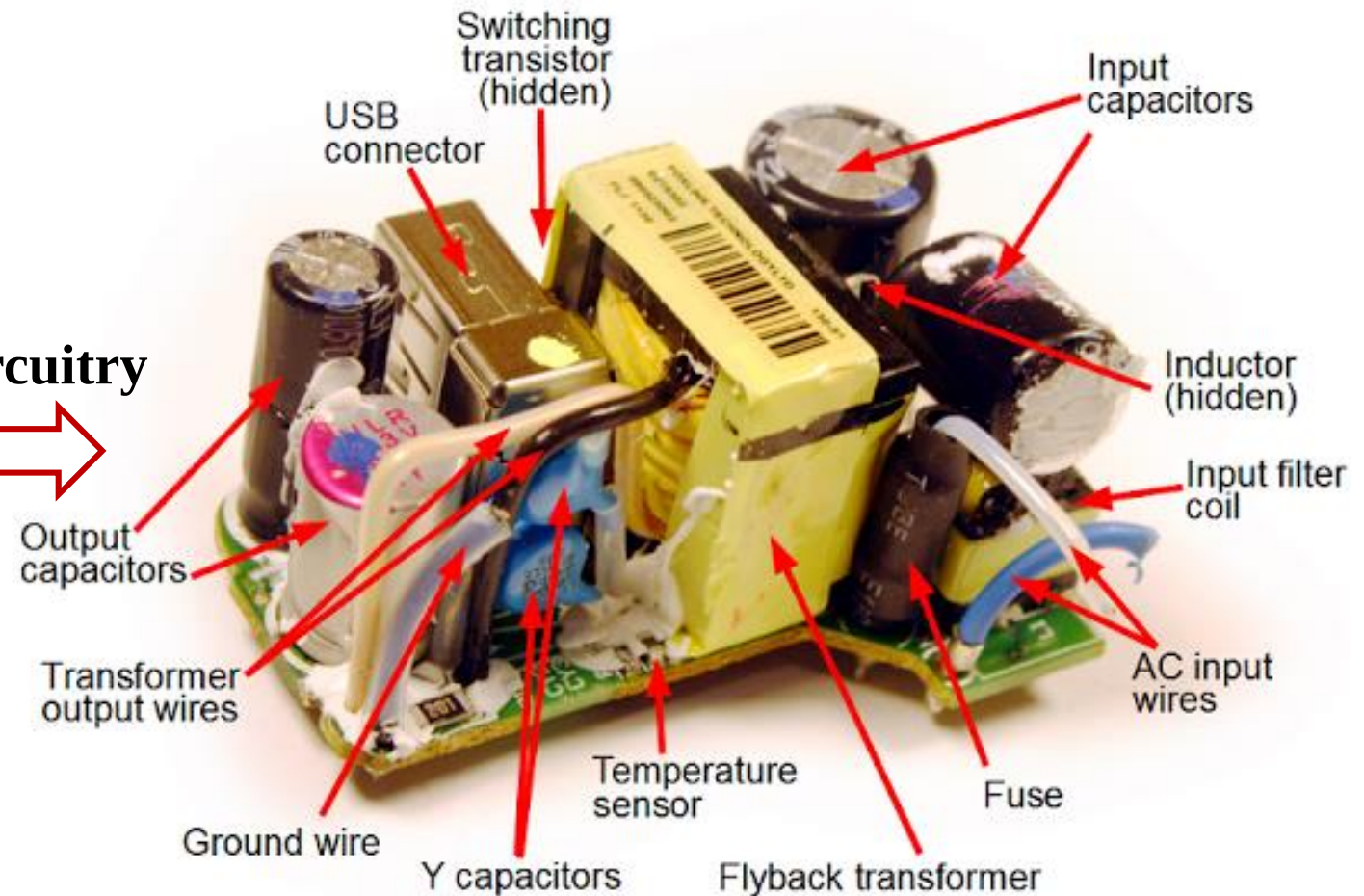


INTRODUCTION

iPad Charger with internal Circuitry



Internal Circuitry

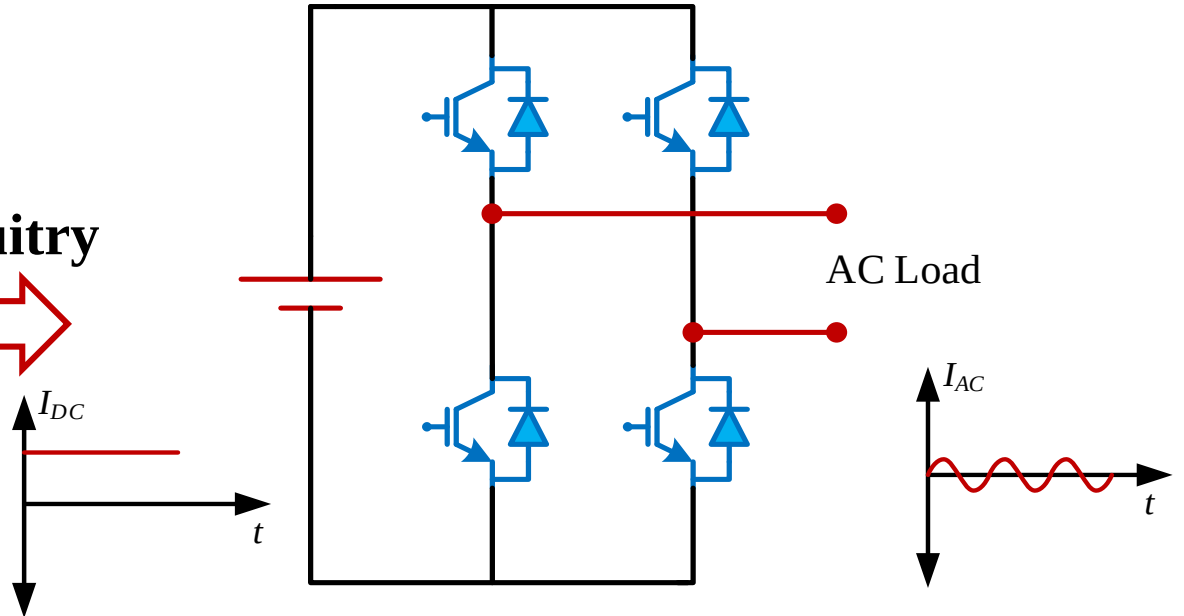


INTRODUCTION

Single Phase Inverter



Internal Circuitry

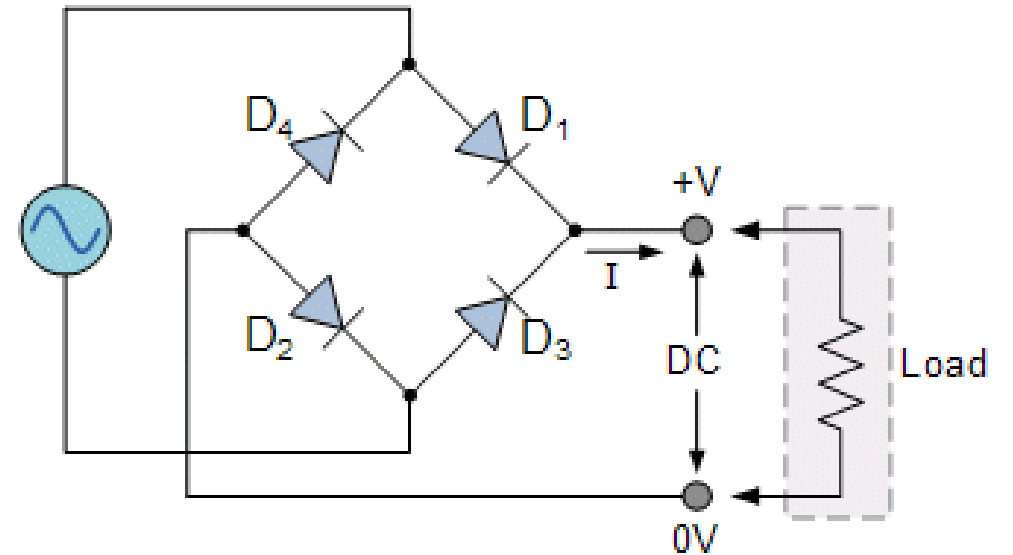


INTRODUCTION

Single phase diode bridge rectifier

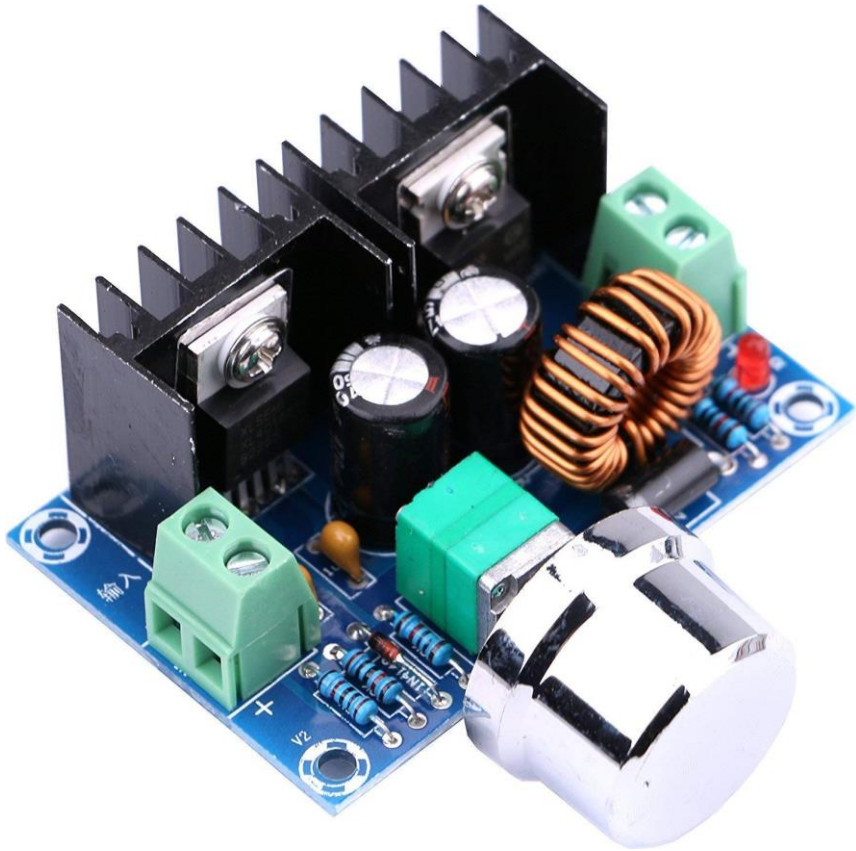


Equivalent Circuit

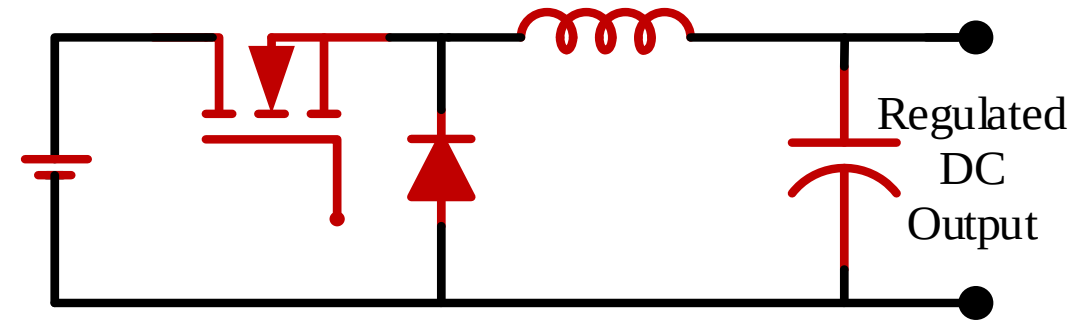


INTRODUCTION

DC-DC step down converter

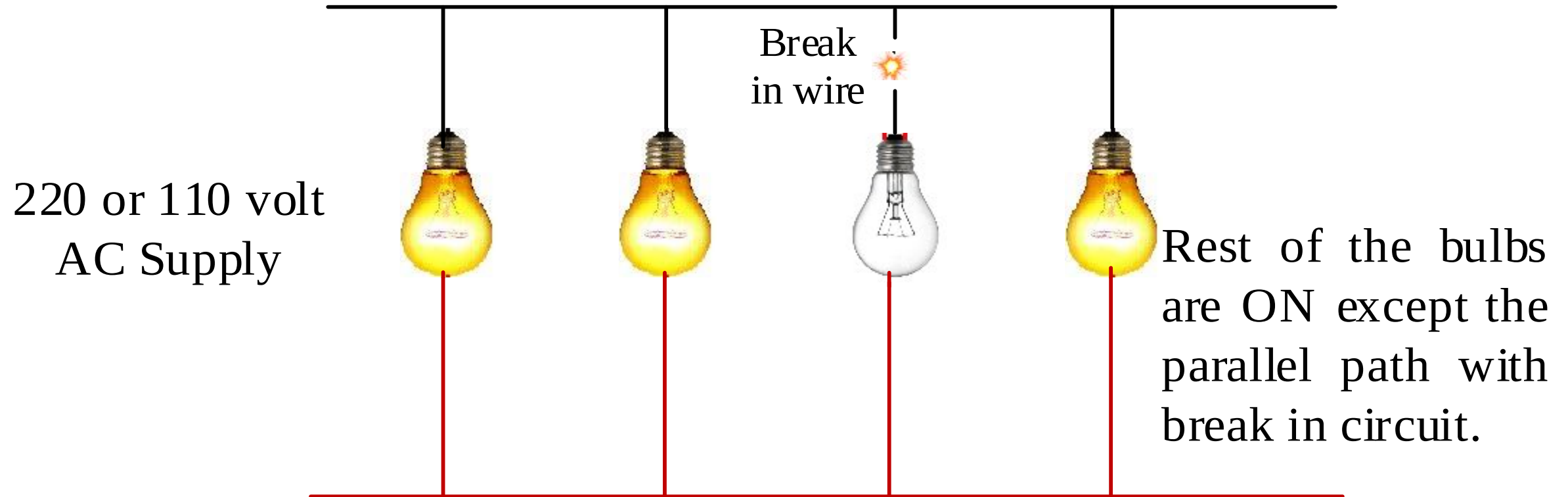


Equivalent Circuit



INTRODUCTION

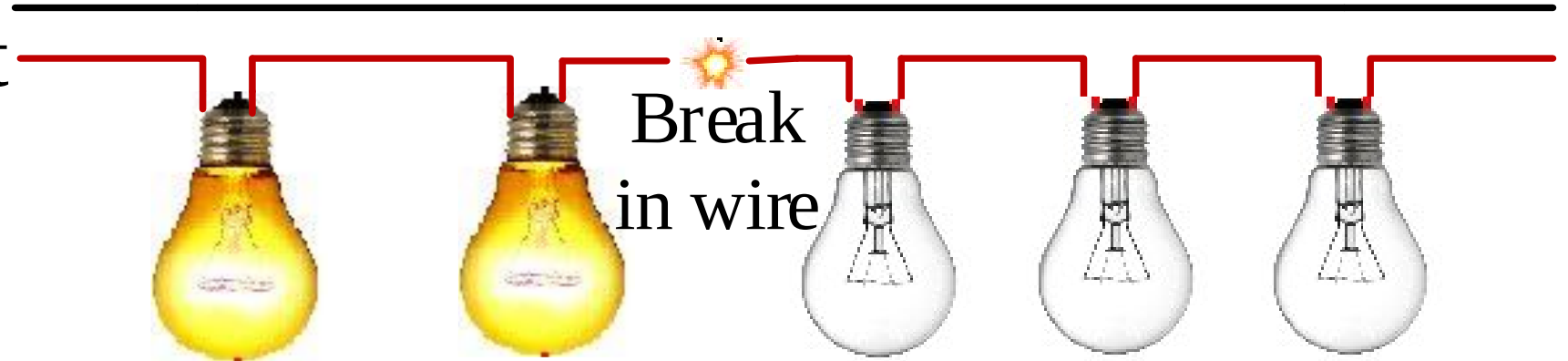
Lighting System Parallel Connection



INTRODUCTION

Lighting System Series Connection

220 or 110 volt
AC Supply

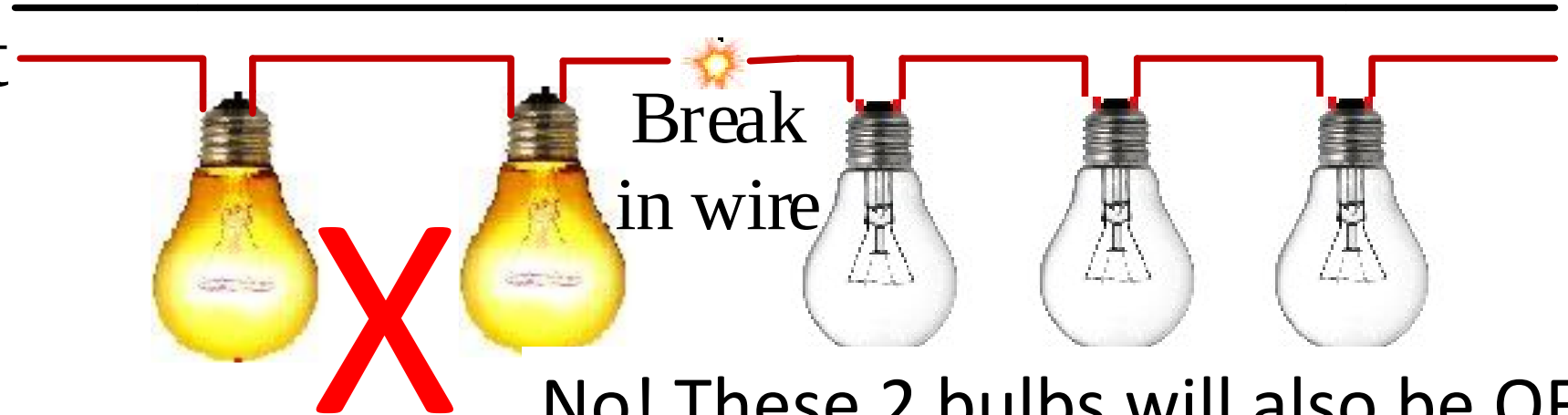


Rest of the bulbs are
OFF after breaking
point. Correct?

INTRODUCTION

Lighting System Series Connection

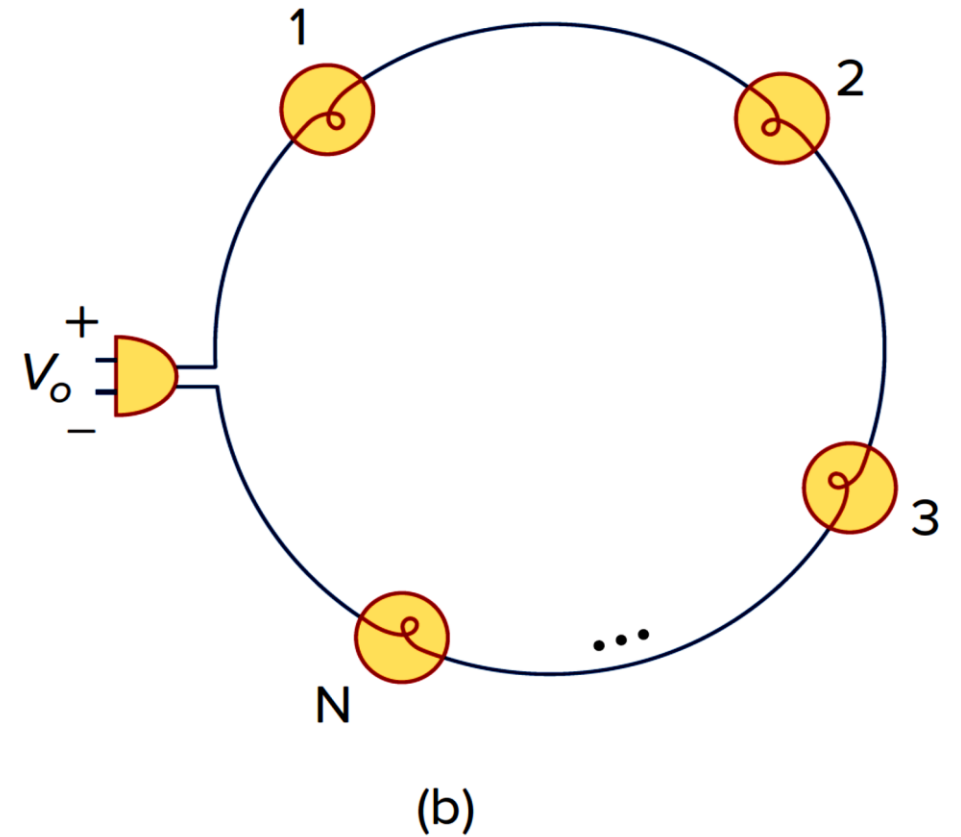
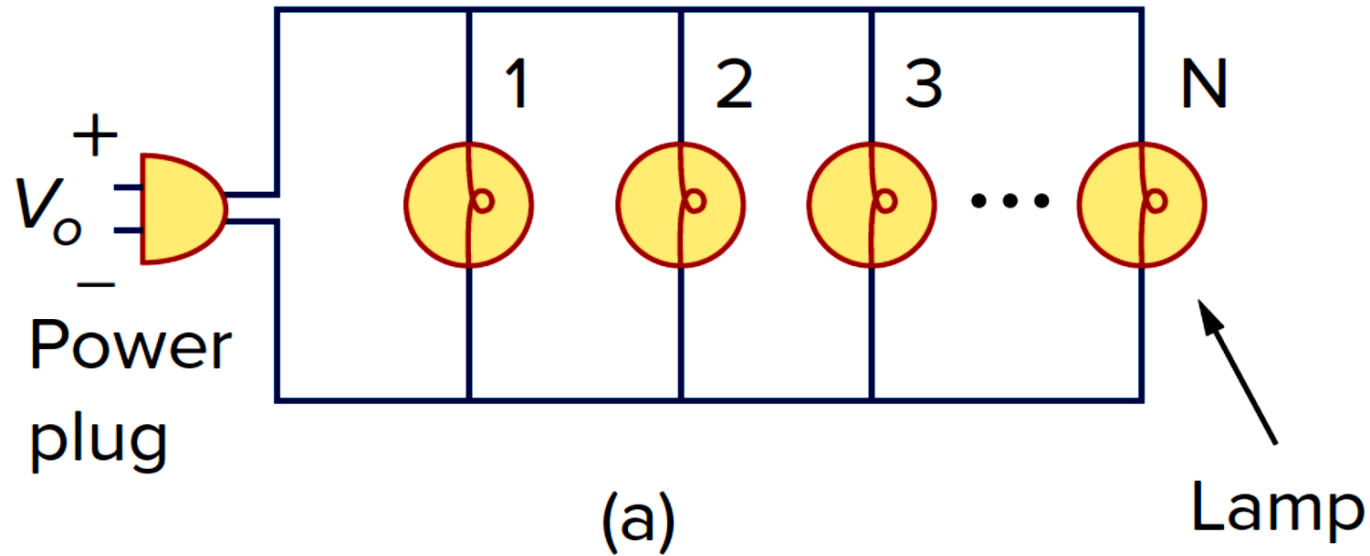
220 or 110 volt
AC Supply



No! These 2 bulbs will also be OFF.
There will be no current on the
entire line because of open circuit.

INTRODUCTION

Lighting System Circuit Representation



(a) Parallel connection of light bulbs, (b) series connection of light bulbs.

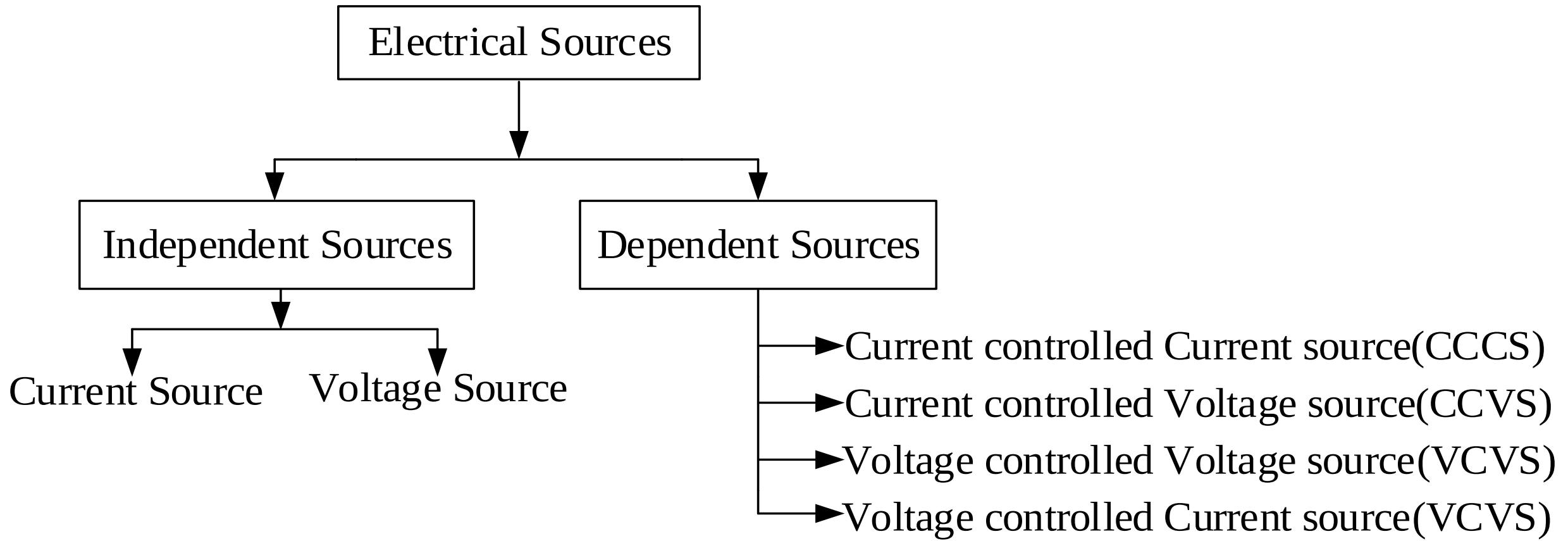
INTRODUCTION

Analysis of Circuits

- ❑ Linear circuit analysis forms the basis for many subsequent electrical engineering courses.
- ❑ The study of electronics relies on the analysis of circuits with devices known as diodes and transistors, which are used to construct power supplies, amplifiers, and digital circuits.
- ❑ The skills which we will develop are typically applied in a rapid, methodical fashion by electronics engineers.

VOLTAGE AND CURRENT SOURCES

Classification



VOLTAGE AND CURRENT SOURCES

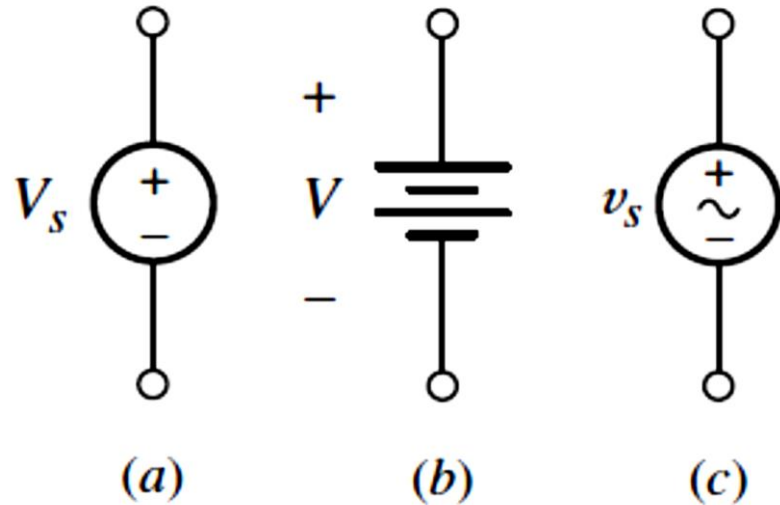
Independent Vs. Dependent Sources

Sources in which the **voltage is completely independent of the current**, or the **current is completely independent of the voltage**; these are termed ***independent sources***.

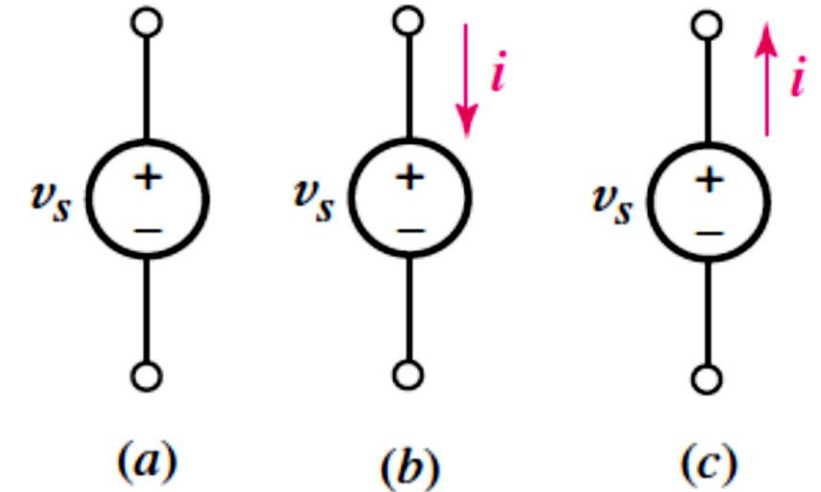
Special kinds of sources for which either the **source voltage or current depends upon a current or voltage elsewhere in the circuit**; such sources are referred to as ***dependent sources***.

VOLTAGE AND CURRENT SOURCES

Independent Voltage Sources



(a) DC voltage source; (b) battery symbol;
(c) AC voltage source symbol



Circuit symbol of the independent voltage source.

An independent voltage source is characterized by a terminal voltage which is completely independent of the current through it.

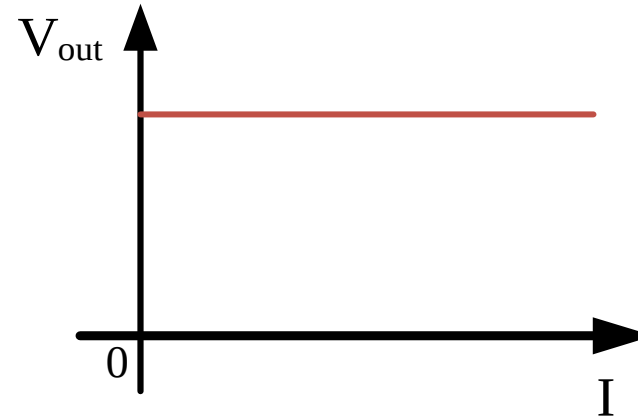
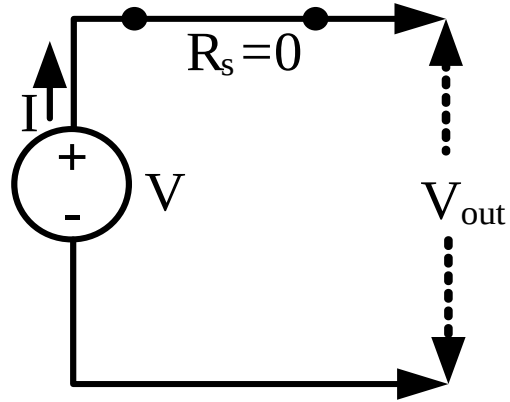
VOLTAGE AND CURRENT SOURCES

Independent Voltage Sources

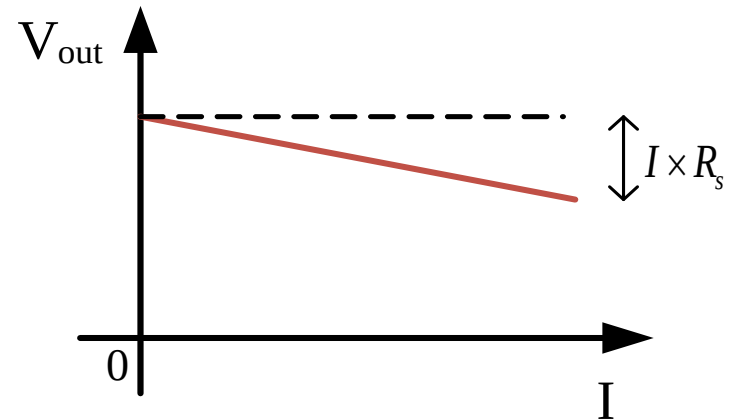
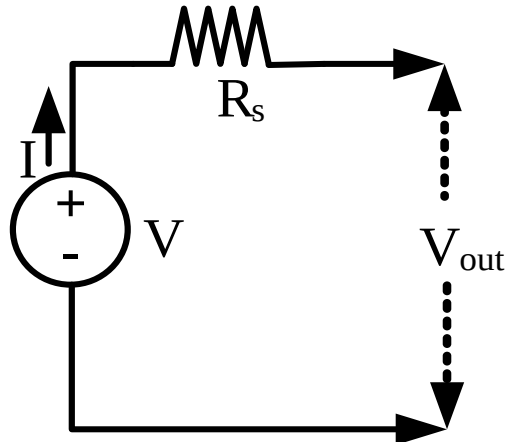
- The independent voltage source is an ideal source and does not represent exactly any real physical device.
- Because the ideal source could theoretically deliver an infinite amount of energy from its terminals (delivery of infinite current at a particular constant voltage).
- This idealized voltage source does, however, furnish a reasonable approximation to several practical voltage sources.

VOLTAGE AND CURRENT SOURCES

Ideal Vs. Practical Voltage Sources



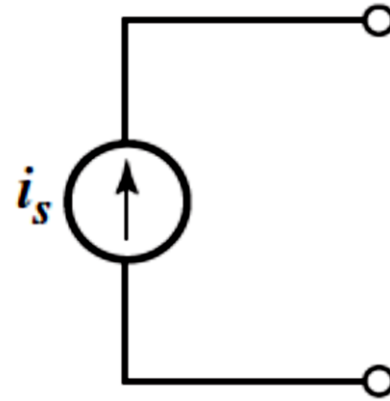
Ideal voltage source equivalent circuit and characteristics



Practical voltage source equivalent circuit and characteristics

VOLTAGE AND CURRENT SOURCES

Independent Current Sources

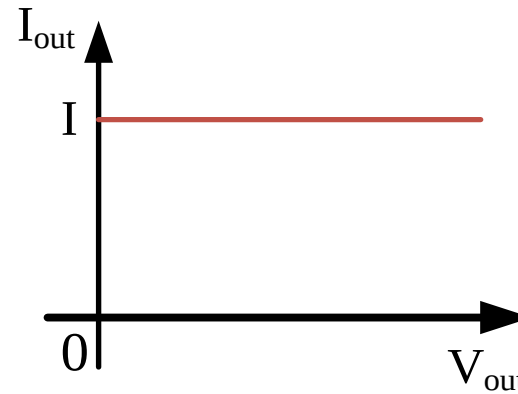
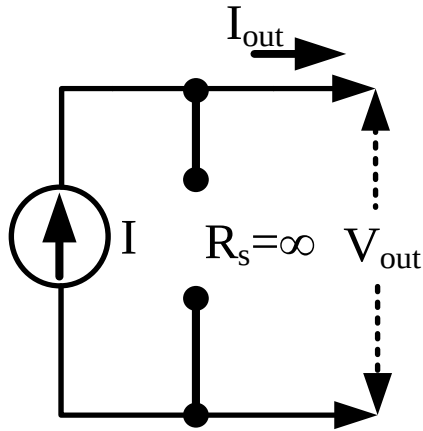


Circuit symbol for the independent current source.

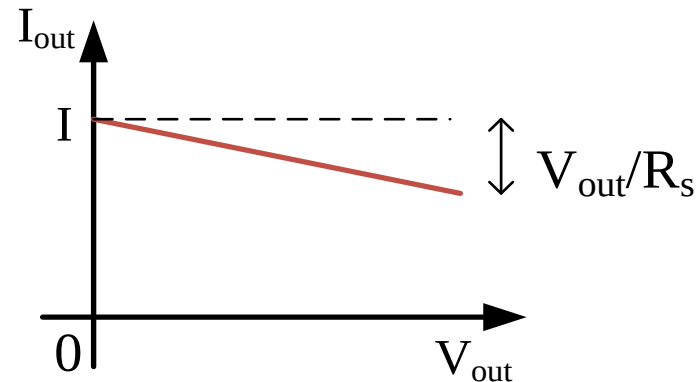
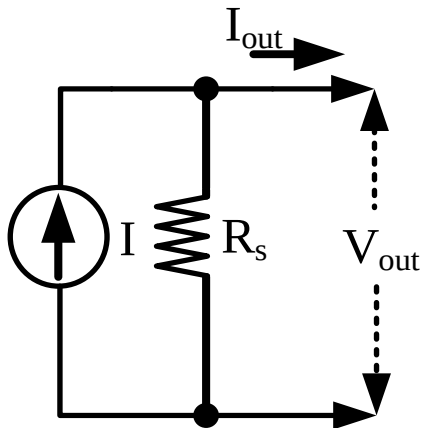
For independent current source, the current through the element is completely independent of the voltage across it.

VOLTAGE AND CURRENT SOURCES

Ideal Vs. Practical Current Sources



Ideal Current source equivalent circuit and characteristics

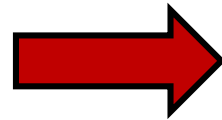


Practical Current source equivalent circuit and characteristics

VOLTAGE AND CURRENT SOURCES

Voltage and Current Source Examples

Voltage Sources

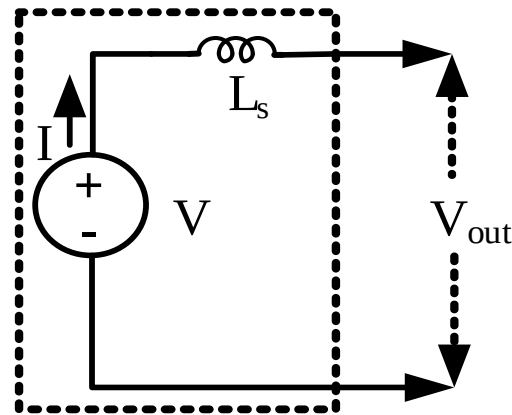


Battery



Diesel Generator

Current Sources



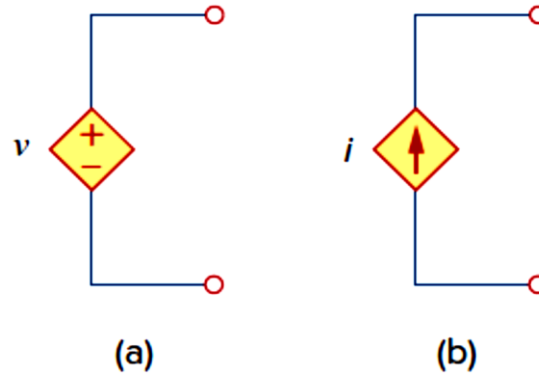
Voltage source with large inductor in series.



Solar cell can act as current source upto certain extent.

VOLTAGE AND CURRENT SOURCES

Dependent Sources

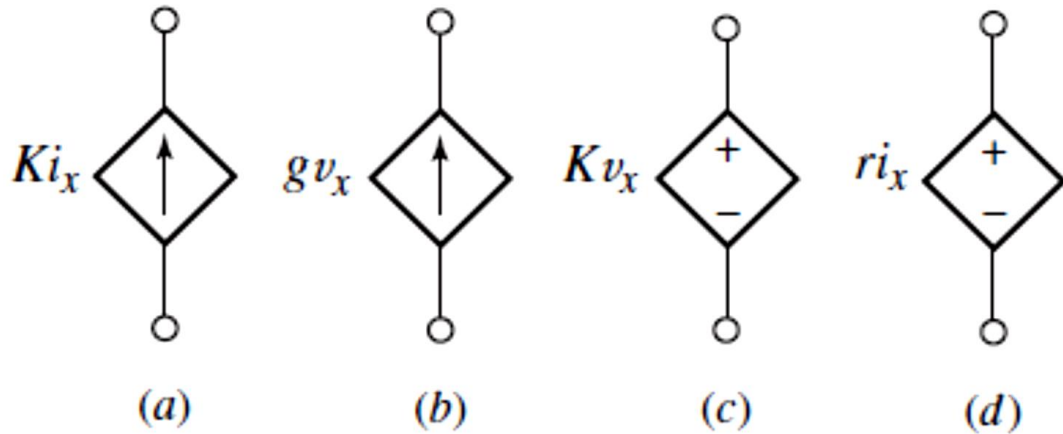


Symbols for: (a) dependent voltage source, (b) dependent current source.

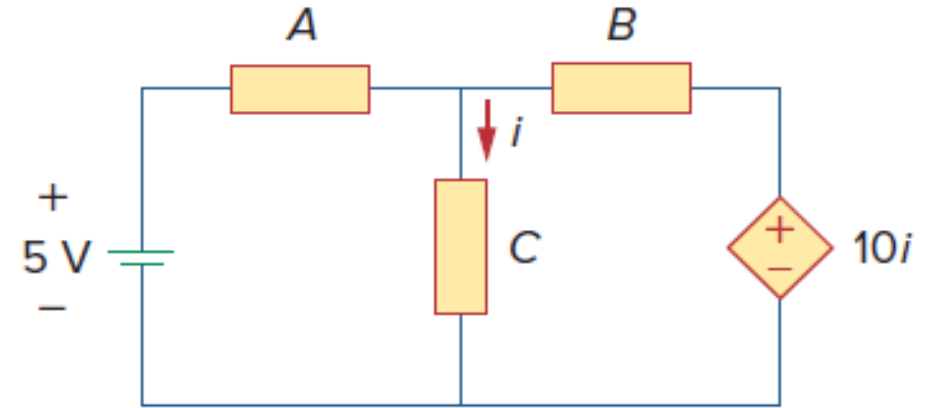
- Dependent sources are useful in modelling elements such as transistors, operational amplifiers, and integrated circuits.
- Dependent sources are usually designated by diamond-shaped symbols.
- Voltage source comes with polarities (+ -) in its symbol, while a current source comes with an arrow, irrespective of what it depends on.

VOLTAGE AND CURRENT SOURCES

Dependent Sources

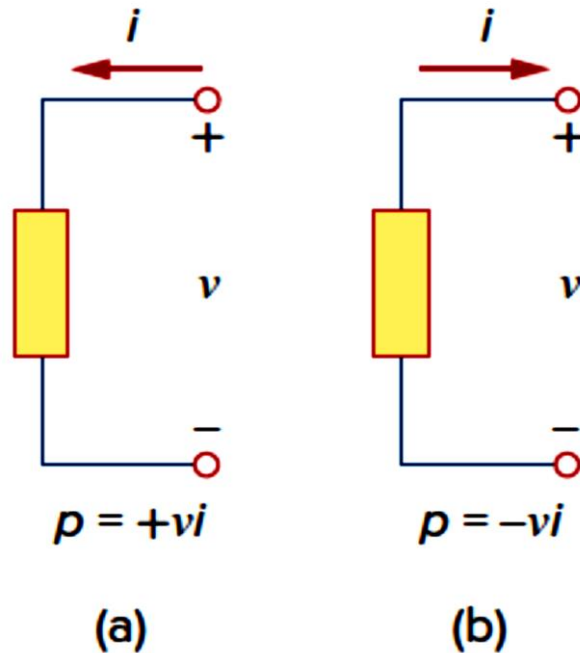


The four different types of dependent sources:
(a) current-controlled current source;
(b) voltage-controlled current source;
(c) voltage-controlled voltage source;
(d) current controlled voltage source.



The source on the right-hand side is a current-controlled voltage source.

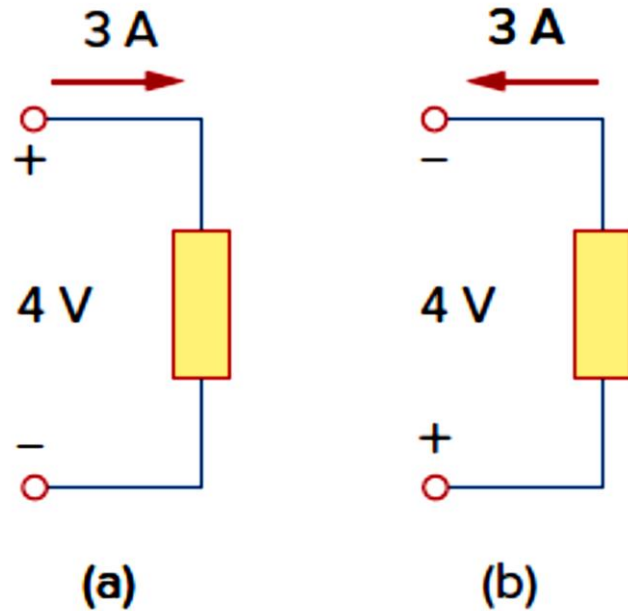
PASSIVE SIGN CONVENTION



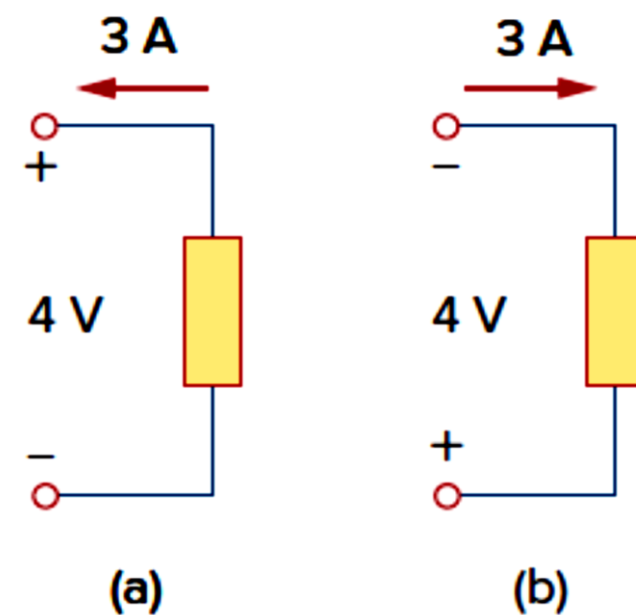
Reference polarities for power using the passive sign convention:
(a) absorbing power, (b) supplying power.

According to the passive sign convention, power assumes a positive sign when the current enters the positive polarity of the voltage across an element.

PASSIVE SIGN CONVENTION



Two cases of an element with an absorbing power of 12 W: (a) $p = 4 \times 3 = 12$ W, (b) $p = 4 \times 3 = 12$ W.



Two cases of an element with a supplying power of 12 W: (a) $p = -4 \times 3 = -12$ W, (b) $p = -4 \times 3 = -12$ W.

PASSIVE SIGN CONVENTION

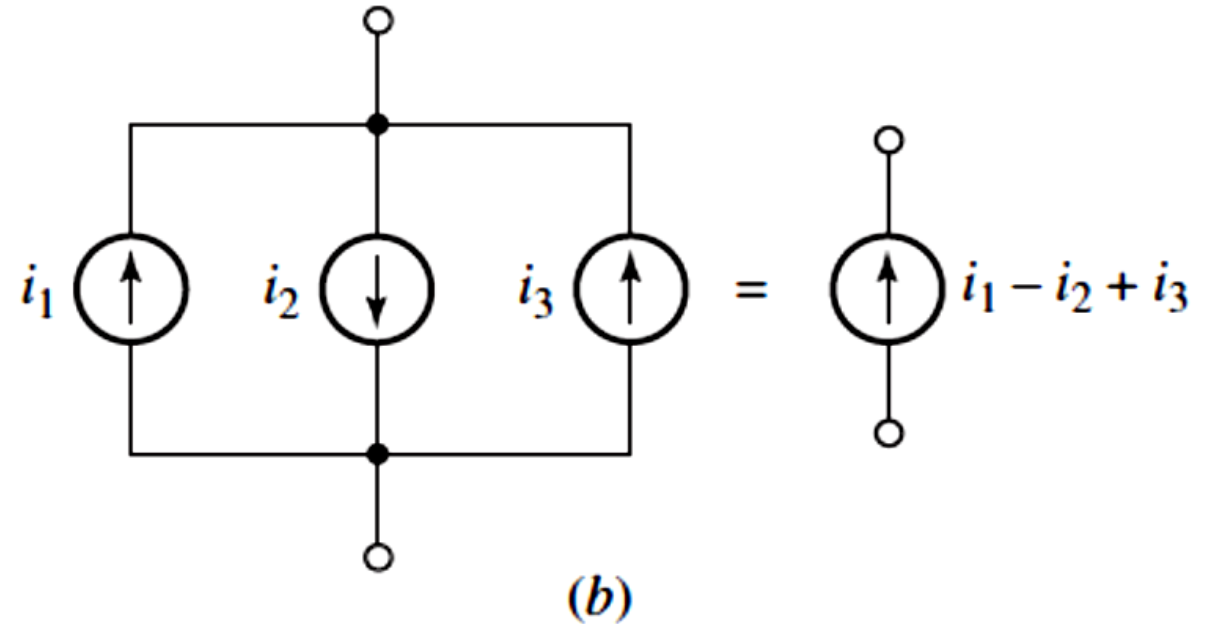
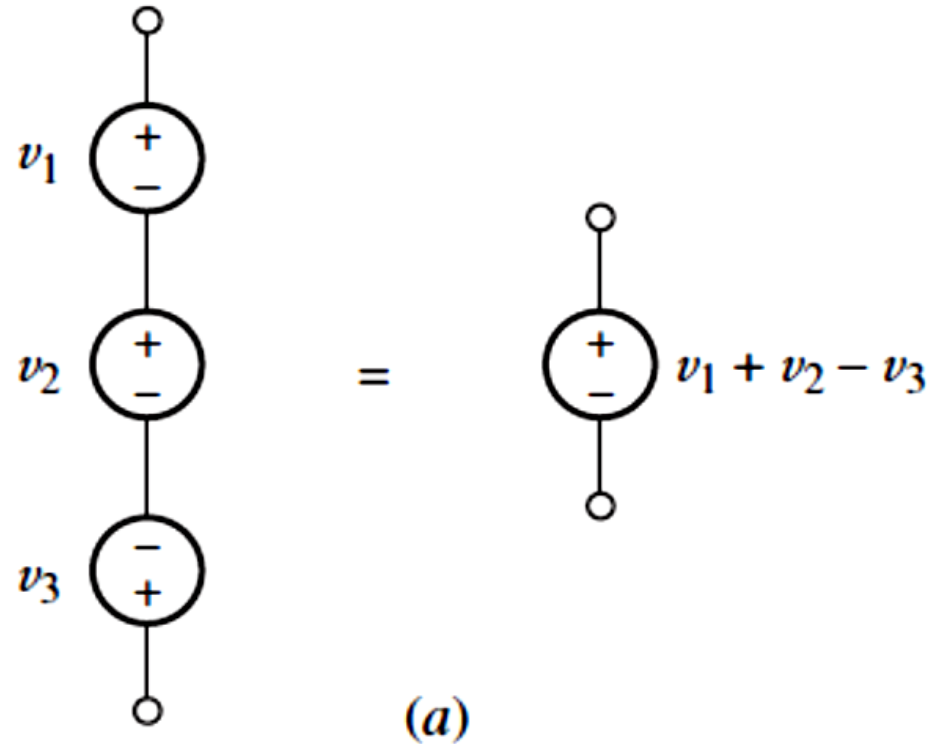
In general,

Power absorbed = Power supplied

- In fact, the *law of conservation of energy* must be obeyed in any electric circuit. For this reason, the algebraic sum of power in a circuit, at any instant of time, must be zero:

$$\sum p = 0$$

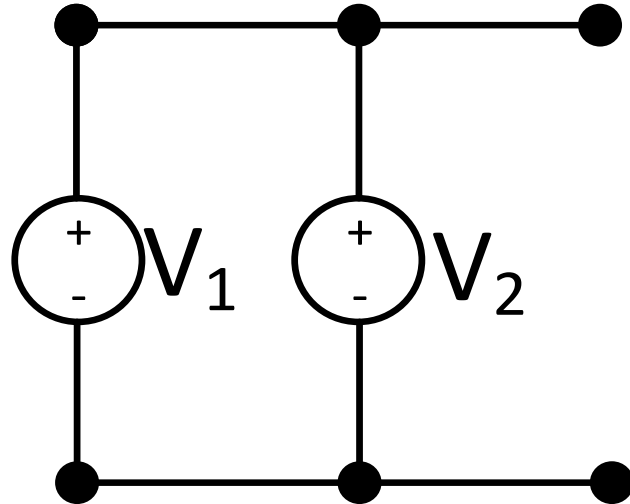
SERIES AND PARALLEL CONNECTION OF SOURCES



(a) Series-connected voltage sources can be replaced by a single source. (b) Parallel current sources can be replaced by a single source.

SERIES AND PARALLEL CONNECTION OF SOURCES

Voltage Sources in Parallel



$$V_1 = V_2$$



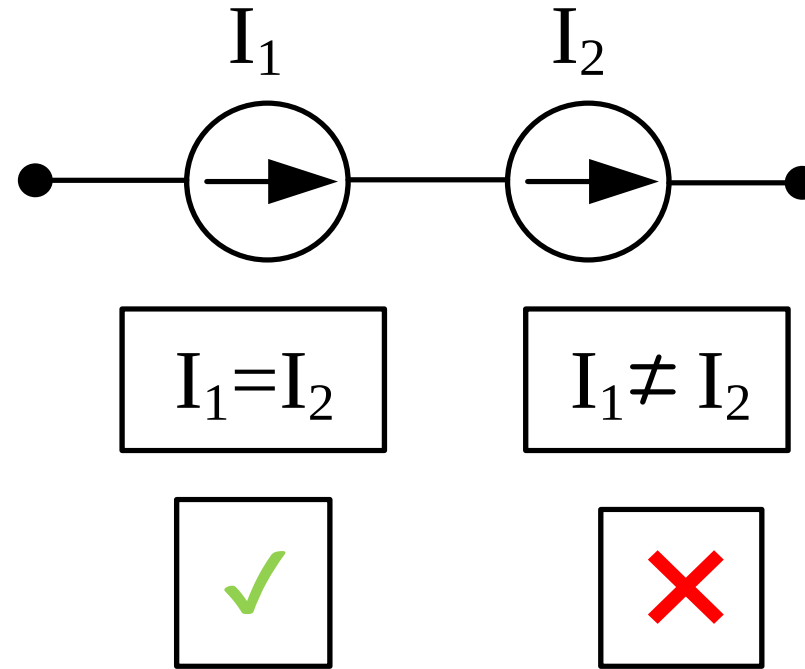
$$V_1 \neq V_2$$



- Two voltage sources are connected in parallel to increase current rating.
- Two unequal voltage sources parallel connection should be avoided as it leads to circulating current among the sources.

SERIES AND PARALLEL CONNECTION OF SOURCES

Current Sources in Series



- Two different current sources must not be connected in series as it violates Kirchhoff's current law.

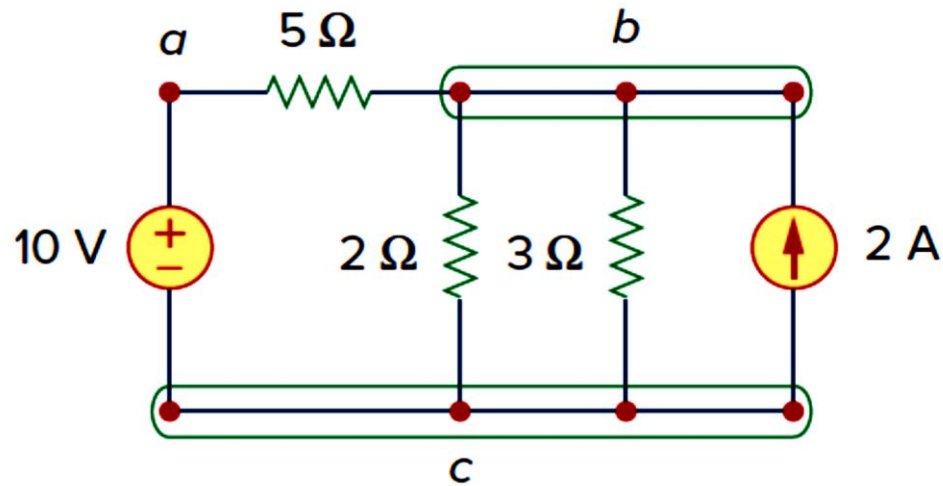
NODE , BRANCHES AND LOOPS

A **branch** represents a single element such as a voltage source or a resistor in a circuit.

A **node** is the point of connection between two or more branches.

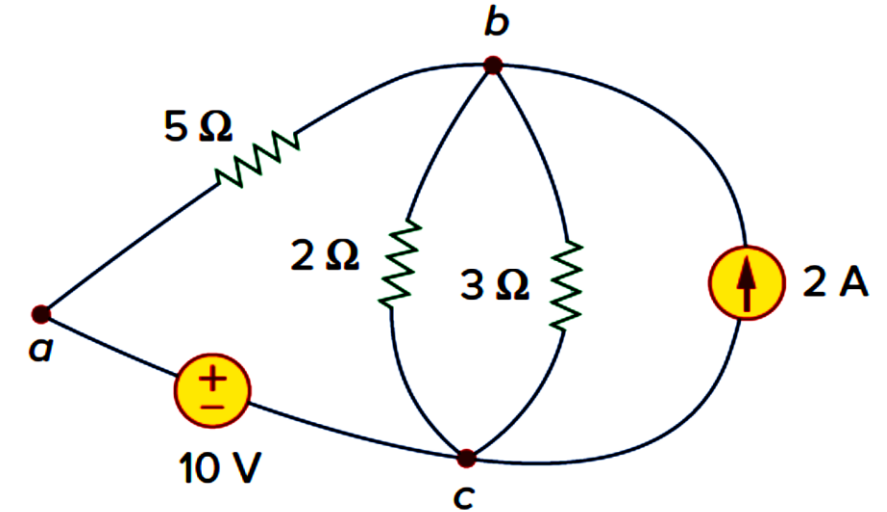
A **loop** is any closed path in a circuit.

NODE , BRANCHES AND LOOPS



5 branches and 3 nodes

Same Circuit
reconfigured



A network with b branches, n nodes, and l independent loops will satisfy the fundamental theorem of network topology:

$$b = l + n - 1$$

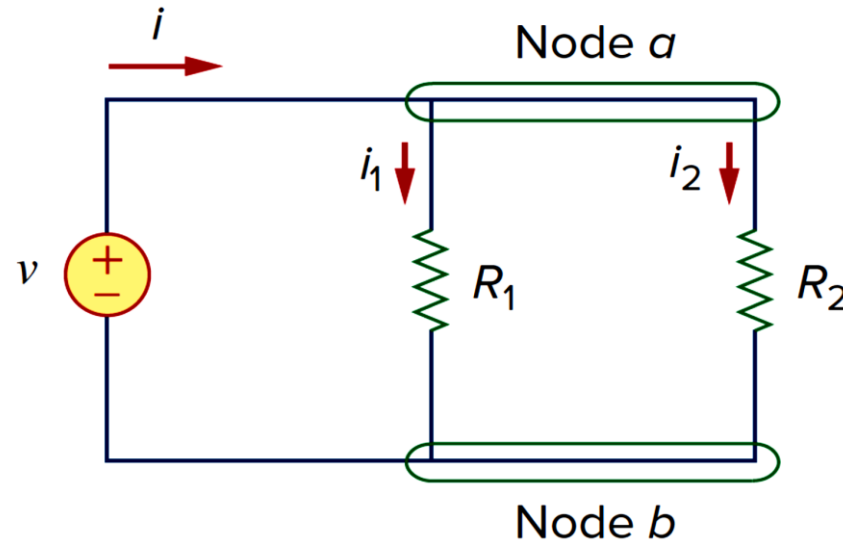
SERIES AND PARALLEL ELEMENTS

Two or more elements are in series if they exclusively share a single node and consequently carry the same current.

Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.

SERIES AND PARALLEL ELEMENTS

Parallel Resistors and Current Division

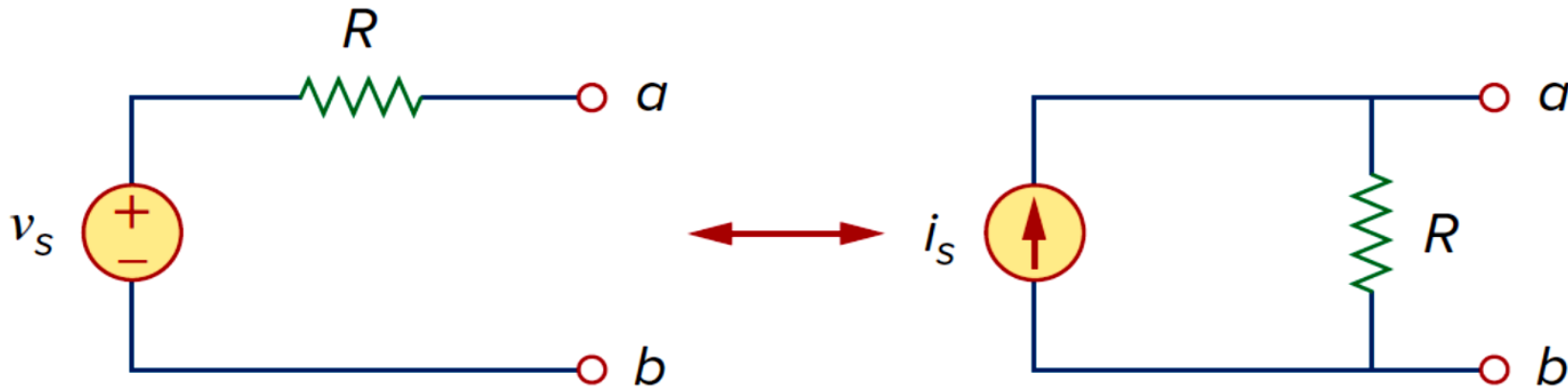


Two Resistor in parallel

$$i_1 = i \times \frac{R_2}{R_1 + R_2}$$

$$i_2 = i \times \frac{R_1}{R_1 + R_2}$$

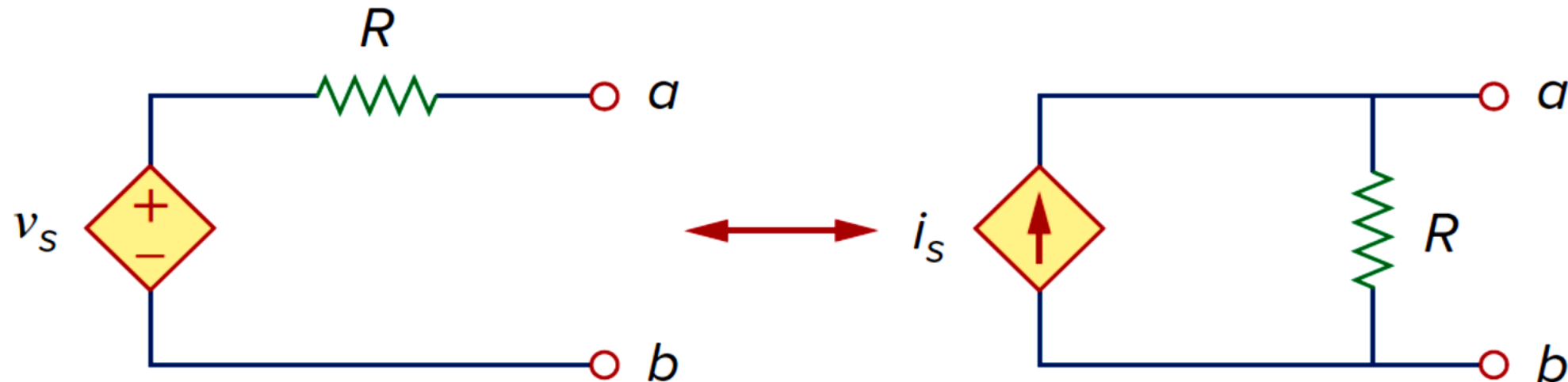
SOURCE TRANSFORMATION



Transformation of independent sources. **Provided**, $v_s = i_s \times R$ or $i_s = \frac{v_s}{R}$

A **source transformation** is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.

SOURCE TRANSFORMATION



Transformation of dependent sources.

- Source transformation also applies to dependent sources.
- A source transformation does not affect the remaining part of the circuit.

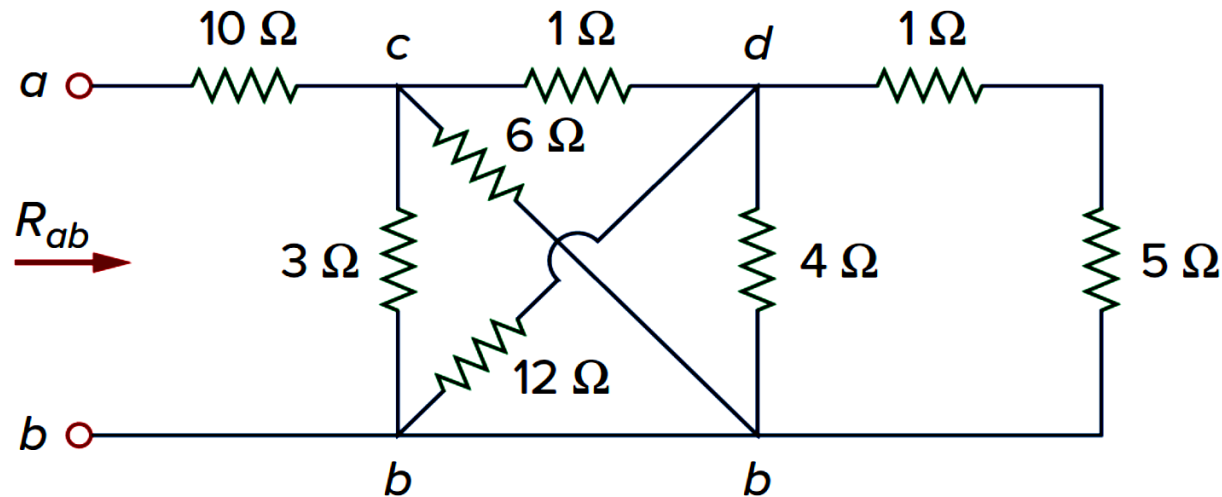
SOURCE TRANSFORMATION

One should keep the following points in mind when dealing with source transformation:

1. The arrow of the current source is directed toward the positive terminal of the voltage source.
2. Source transformation is not possible when $R = 0$, which is the case with an ideal voltage source. However, for a practical, non-ideal voltage source, $R \neq 0$.
3. Similarly, an ideal current source with $R = \infty$ cannot be replaced by a finite voltage source.

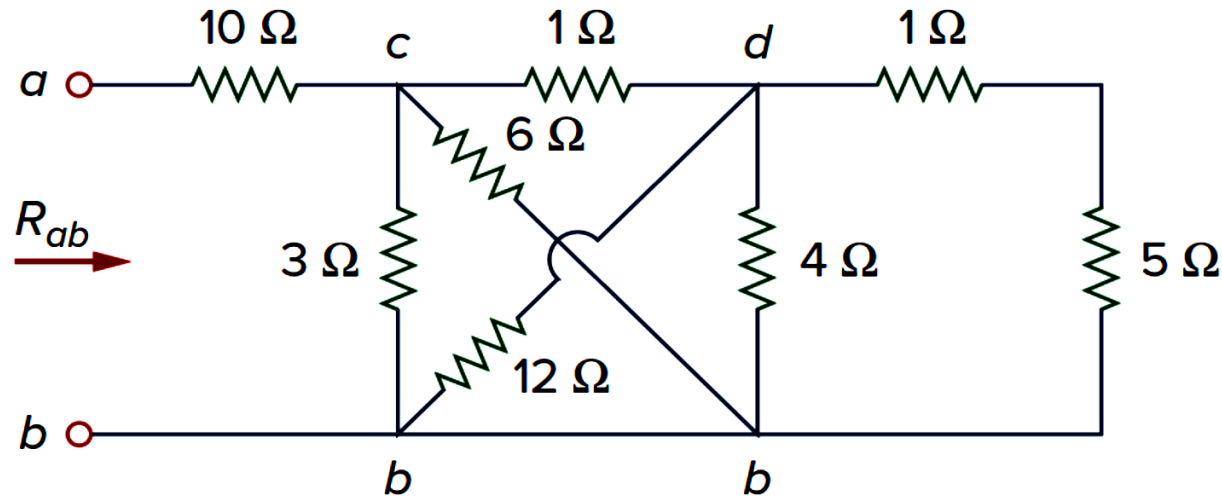
EXERCISE / NUMERICAL ANALYSIS

Q1. Find R_{ab} for the circuit shown in Figure.



EXERCISE / NUMERICAL ANALYSIS

Q1. Find R_{ab} for the circuit shown in Figure.

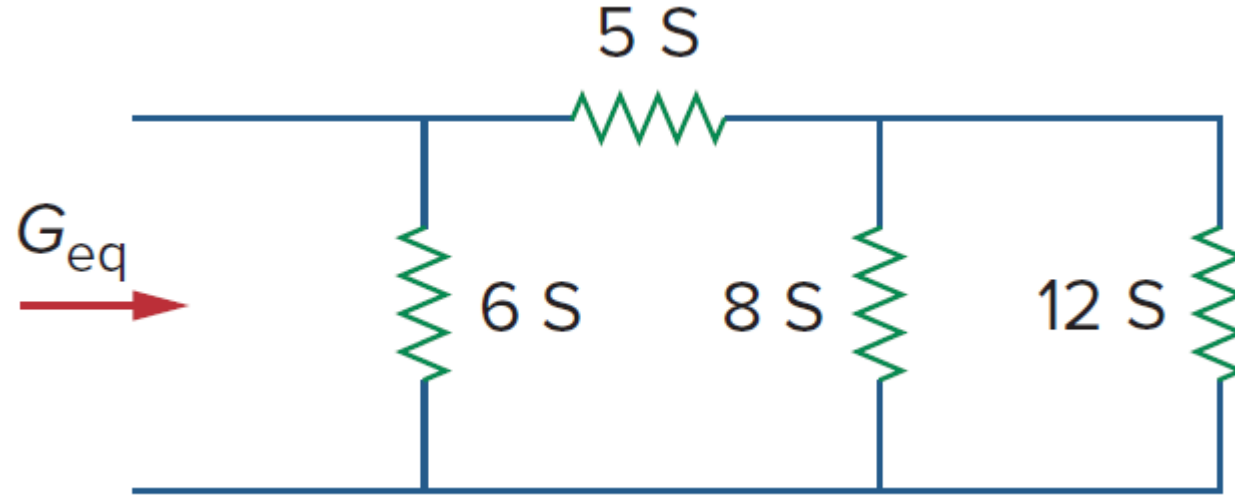


Answer:

$$R_{ab} = \left[\left\{ (4 \parallel 12) \parallel (5 + 1) \right\} + 1 \right] \parallel (6 \parallel 3) + 10 = 11.2\ \Omega$$

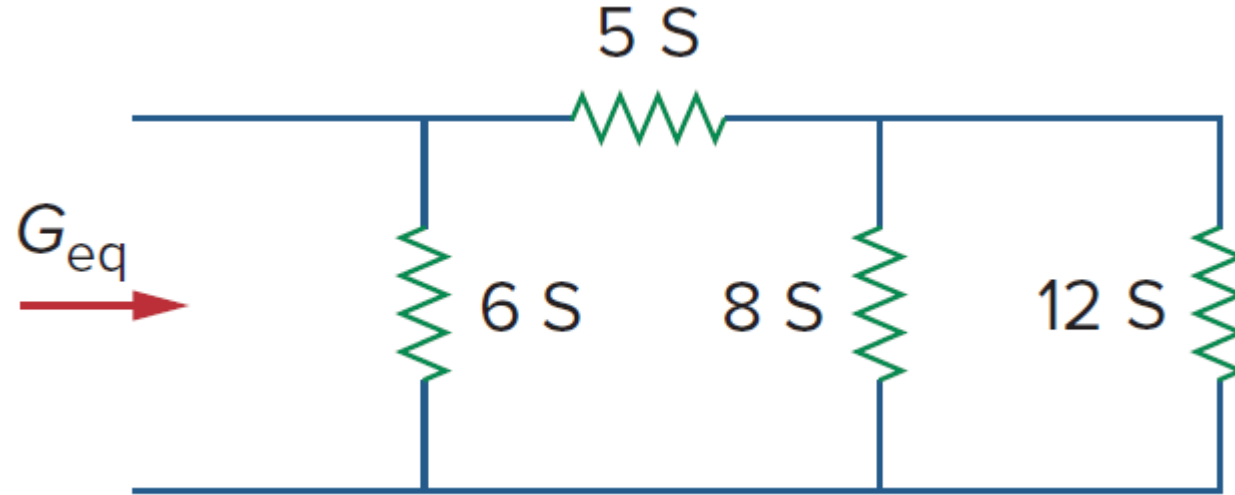
EXERCISE / NUMERICAL ANALYSIS

Q2. Find G_{eq} for the circuit shown in Figure.



EXERCISE / NUMERICAL ANALYSIS

Q2. Find G_{eq} for the circuit shown in Figure.

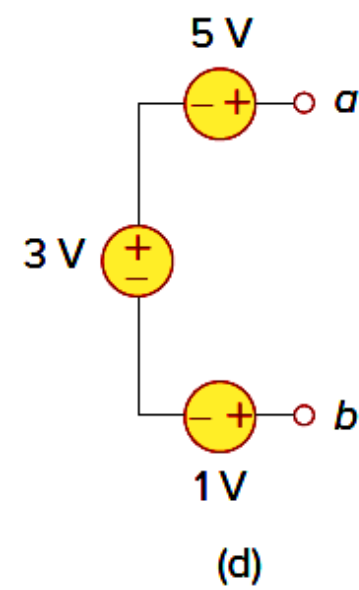
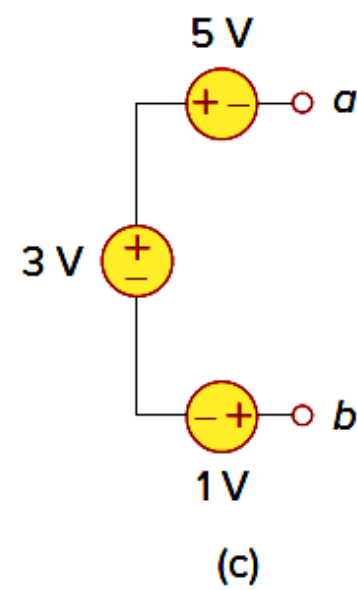
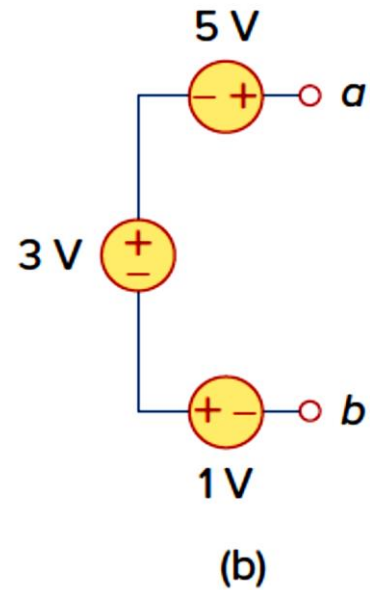
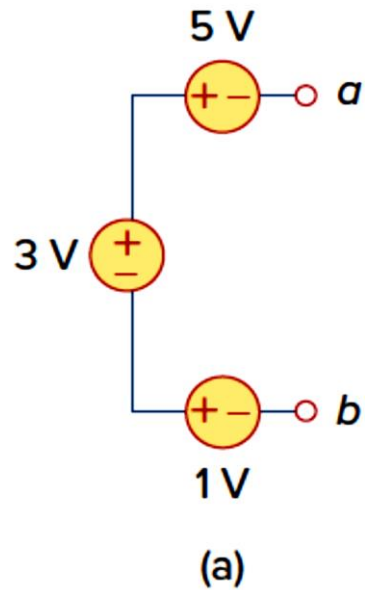


Answer:

$$G_{eq} = \frac{(12 + 8) \times 5}{(12 + 8) + 5} + 6 = 10 \text{ S or } \mathcal{U}$$

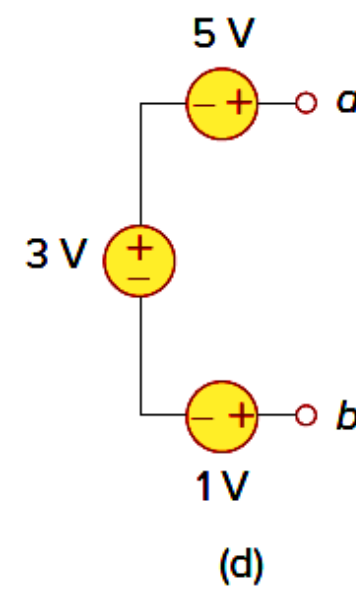
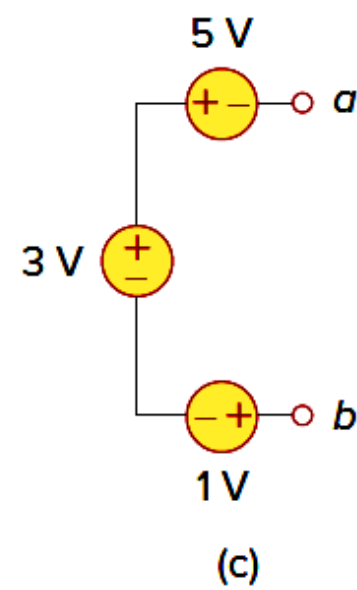
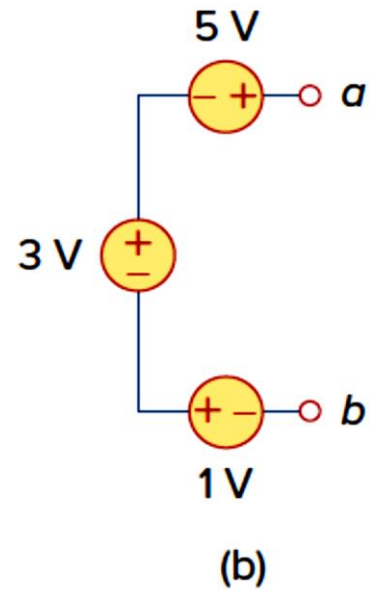
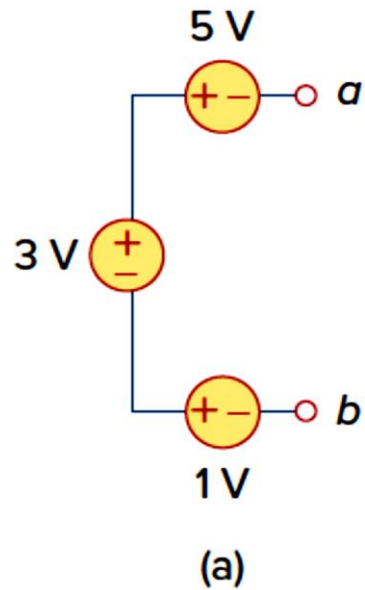
EXERCISE / NUMERICAL ANALYSIS

Q3. Which of the circuits in following figure will give you $V_{ab} = 7 \text{ V}$?



EXERCISE / NUMERICAL ANALYSIS

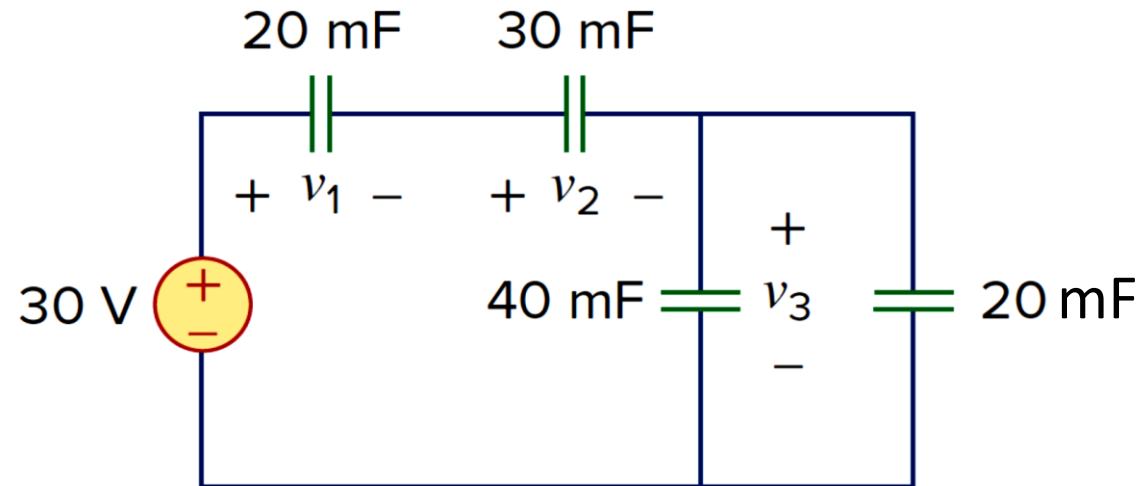
Q3. Which of the circuits in following figure will give you $V_{ab} = 7 \text{ V}$?



Answer: (d)

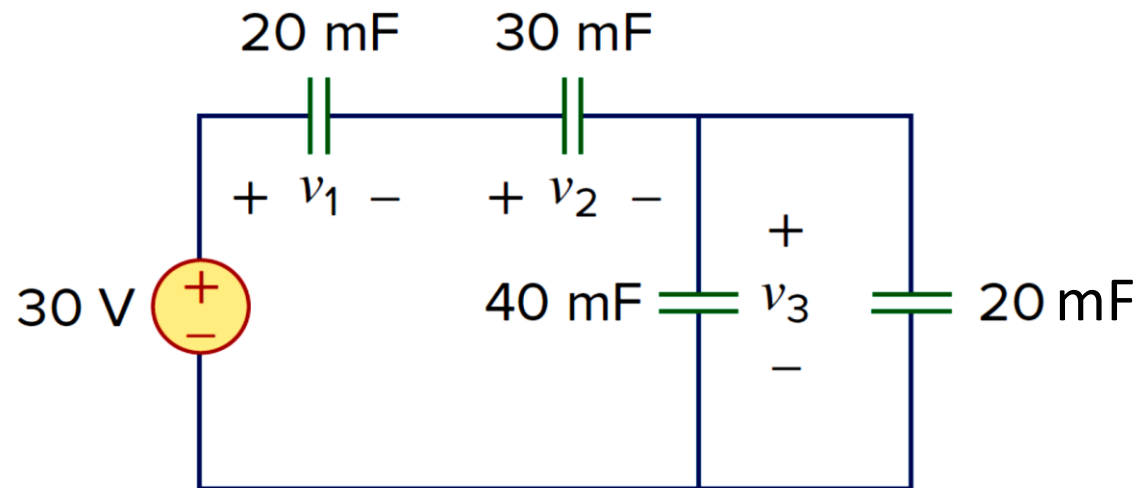
EXERCISE / NUMERICAL ANALYSIS

Q4. For the circuit in figure below, find the voltage across each capacitor.



EXERCISE / NUMERICAL ANALYSIS

Q4. For the circuit in figure below, find the voltage across each capacitor.



Answer:

Here equivalent capacitance is

$$C_{eq} = \frac{1}{\frac{1}{60} + \frac{1}{30} + \frac{1}{20}} = 10 \text{ mF}$$

EXERCISE / NUMERICAL ANALYSIS

The total charge is

$$q = C_{eq} v = 10 \times 10^{-3} \times 30 = 0.3 C$$

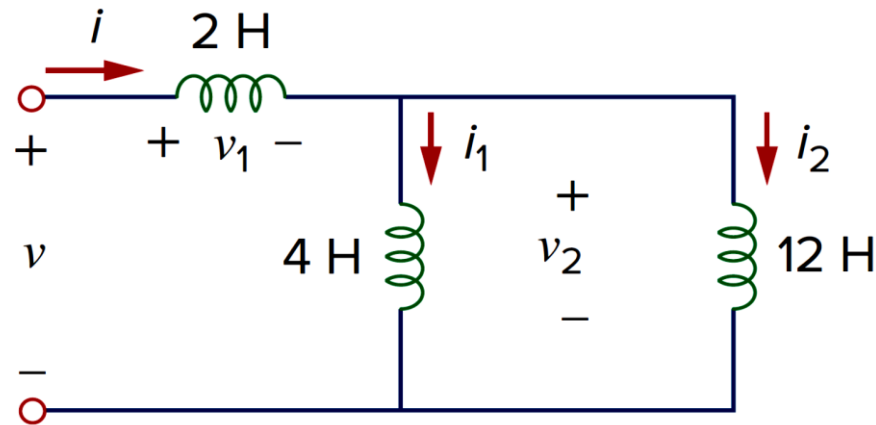
This is the charge on the 20-mF and 30-mF capacitors, because they are in series with the 30-V source. (A crude way to see this is to imagine that charge acts like current, since $i = dq/dt$.) Therefore,

$$v_1 = \frac{q}{C_1} = \frac{0.3}{20 \times 10^{-3}} = 15V; \quad v_2 = \frac{q}{C_2} = \frac{0.3}{30 \times 10^{-3}} = 10V$$

$$\text{Now, } v_3 = 30 - v_1 - v_2 = 30 - 15 - 10 = 5V$$

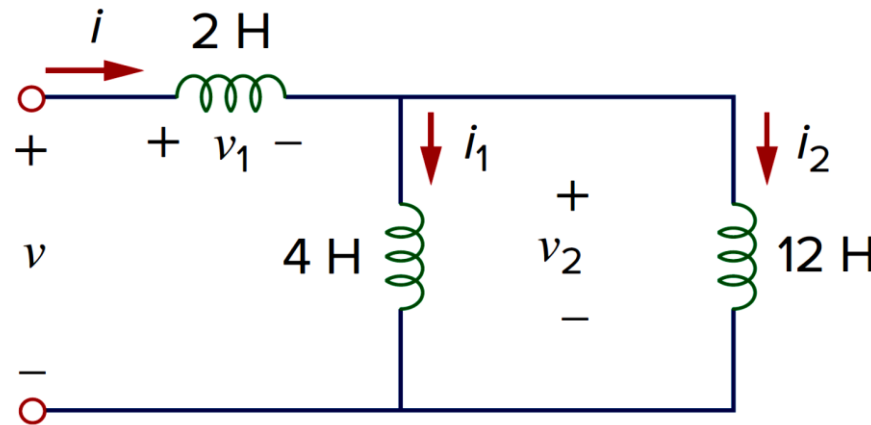
EXERCISE / NUMERICAL ANALYSIS

Q5. For the circuit in below figure, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(0)$, $v_1(0)$, and $v_2(0)$



EXERCISE / NUMERICAL ANALYSIS

Q5. For the circuit in below figure, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(0)$, $v_1(0)$, and $v_2(0)$



Answer:

(a) From $i(t) = 4(2 - e^{-10t})$ mA, $i(0) = 4(2 - 1) = 4$ mA. Since $i = i_1 + i_2$,
 $i_1(0) = i(0) - i_2(0) = 4 - (-1) = 5$ mA

EXERCISE / NUMERICAL ANALYSIS

(b) The equivalent inductance is $L_{eq} = 2 + 4 \parallel 12 = 2 + 3 = 5 \text{ H}$

$$v(t) = L_{eq} \frac{di}{dt} = 5(4)(-1)(-10)e^{-10t} = 200e^{-10t} \text{ mV}$$

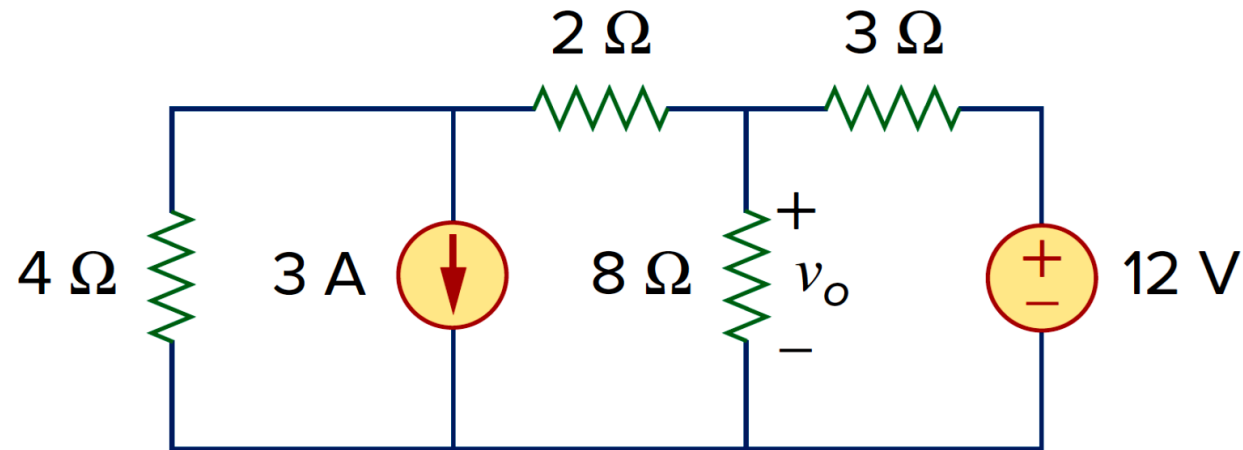
$$v_1(t) = 2 \frac{di}{dt} = 2(-4)(-10)e^{-10t} = 80e^{-10t} \text{ mV}$$

$$v_2(t) = v(t) - v_1(t) = 120e^{-10t} \text{ mV}$$

$$\Rightarrow \quad \mathbf{v(0) = 200 \text{ mV}, } \mathbf{v_1(0) = 80 \text{ mV}, \text{ and } v_2(0) = 120 \text{ mV}}$$

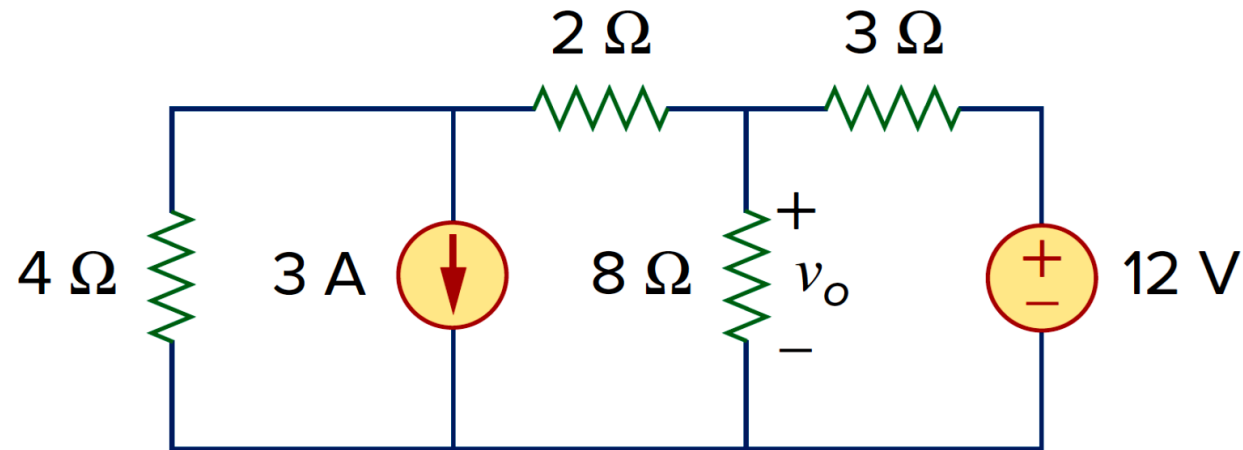
EXERCISE / NUMERICAL ANALYSIS

Q6. Use source transformation to find v_o in the circuit of below figure.



EXERCISE / NUMERICAL ANALYSIS

Q6. Use source transformation to find v_o in the circuit of below figure.

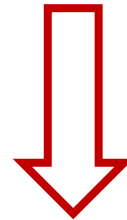
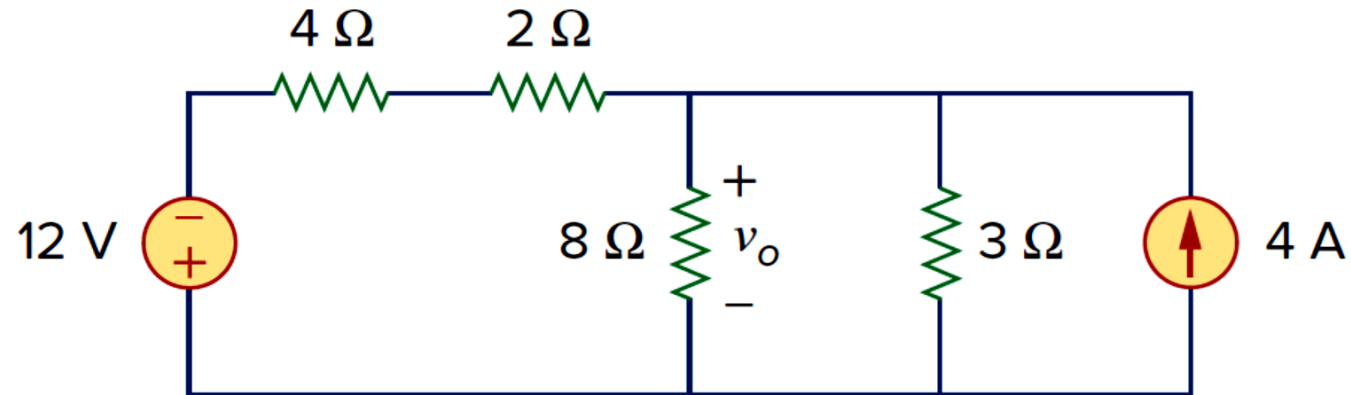


Answer:

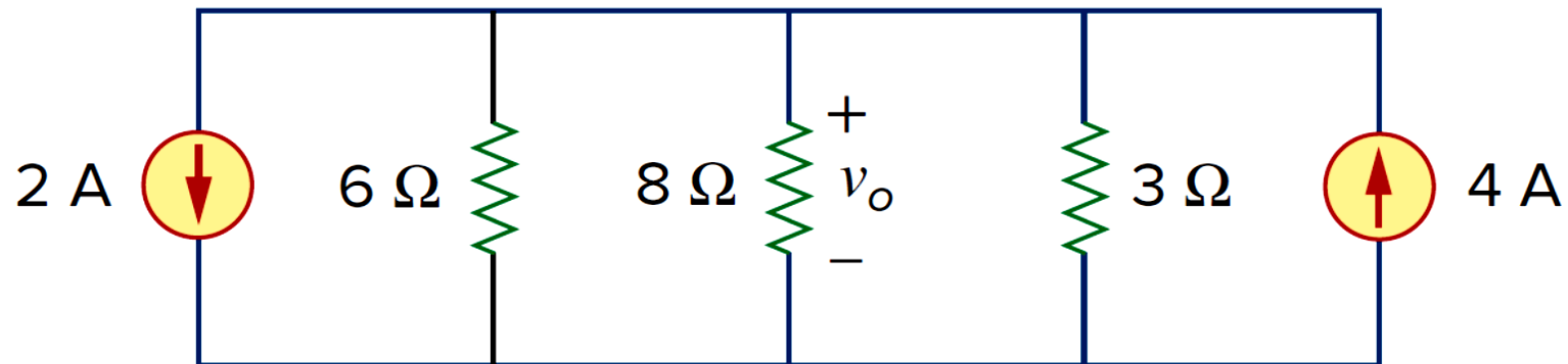
Converting 3 A current source in parallel with $4\ \Omega$ into a voltage source will bring $4\ \Omega$ and $2\ \Omega$ in series.

Converting 12 V voltage source in series with $3\ \Omega$ into a current source will bring $3\ \Omega$ and $8\ \Omega$ in parallel.

EXERCISE / NUMERICAL ANALYSIS

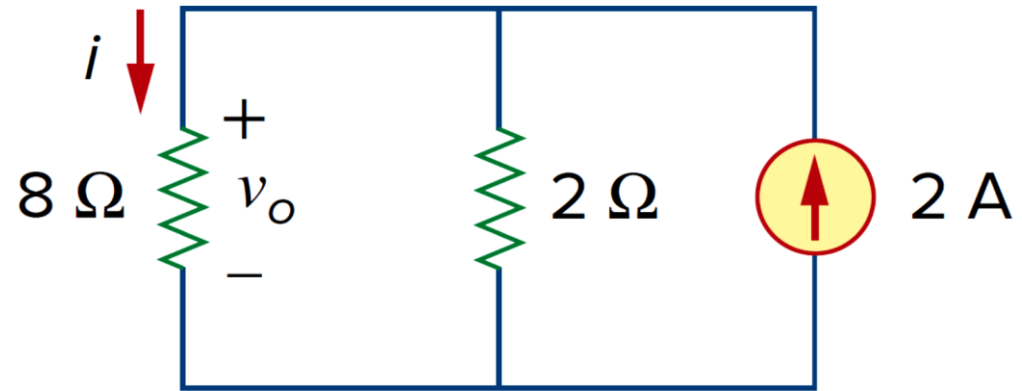


Converting 12V voltage source
again into a current source



EXERCISE / NUMERICAL ANALYSIS

Finally all elements are in parallel and equivalent circuit reduces to



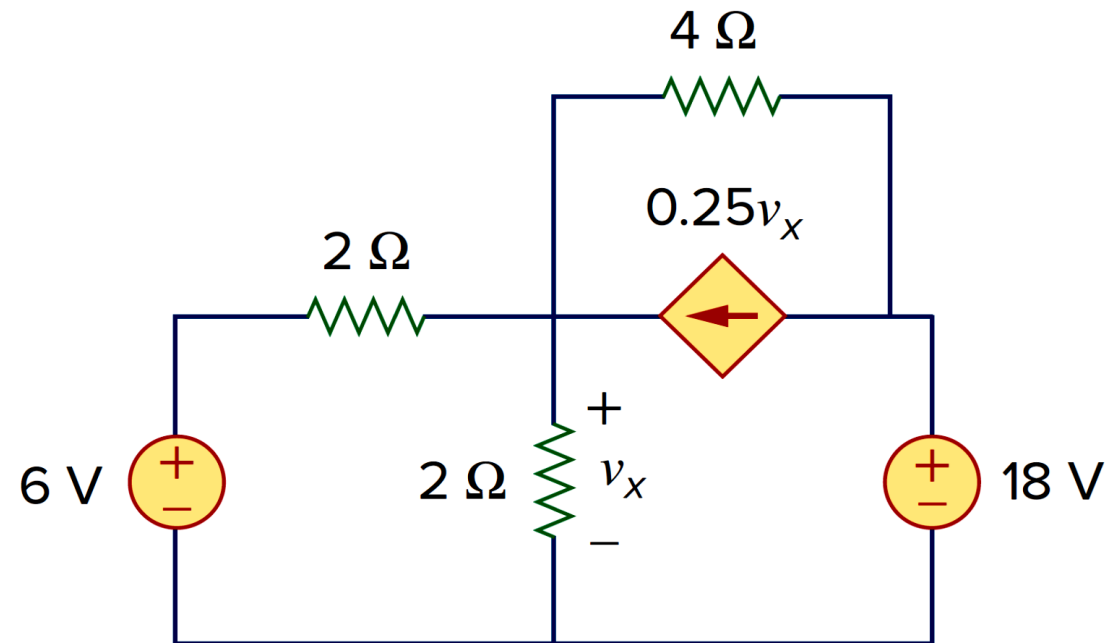
Use current division to get

$$i = 2 \times \frac{2}{2 + 8} = 0.4\text{ A}$$

$$v_o = 8i = 8(0.4) = 3.2\text{ V}$$

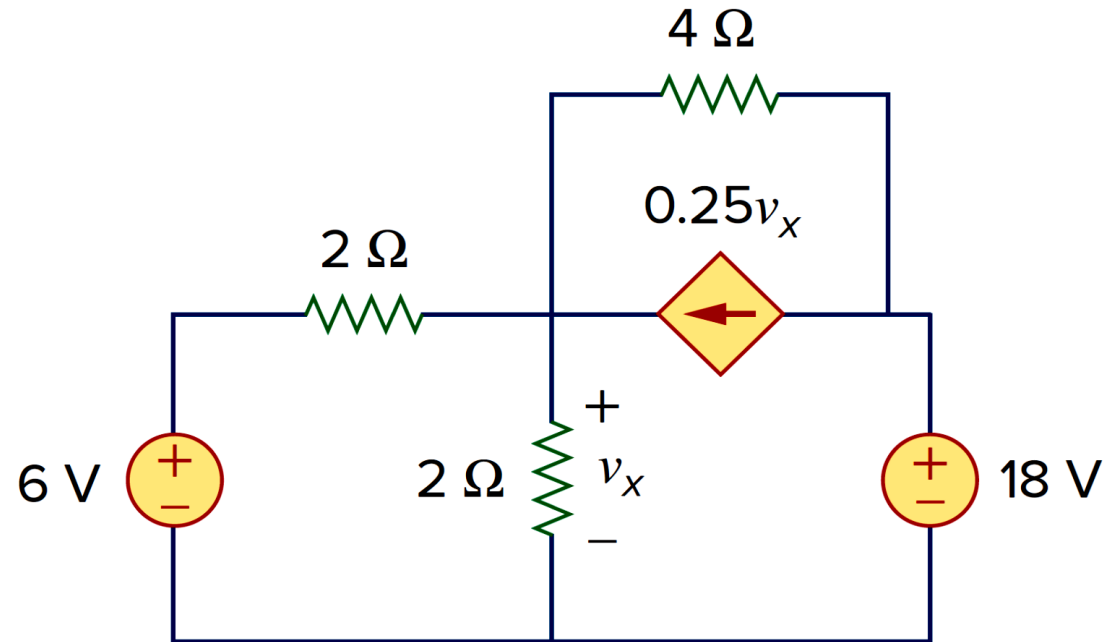
EXERCISE / NUMERICAL ANALYSIS

Q7. Find v_x in the given figure using source transformation.



EXERCISE / NUMERICAL ANALYSIS

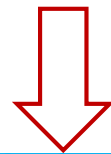
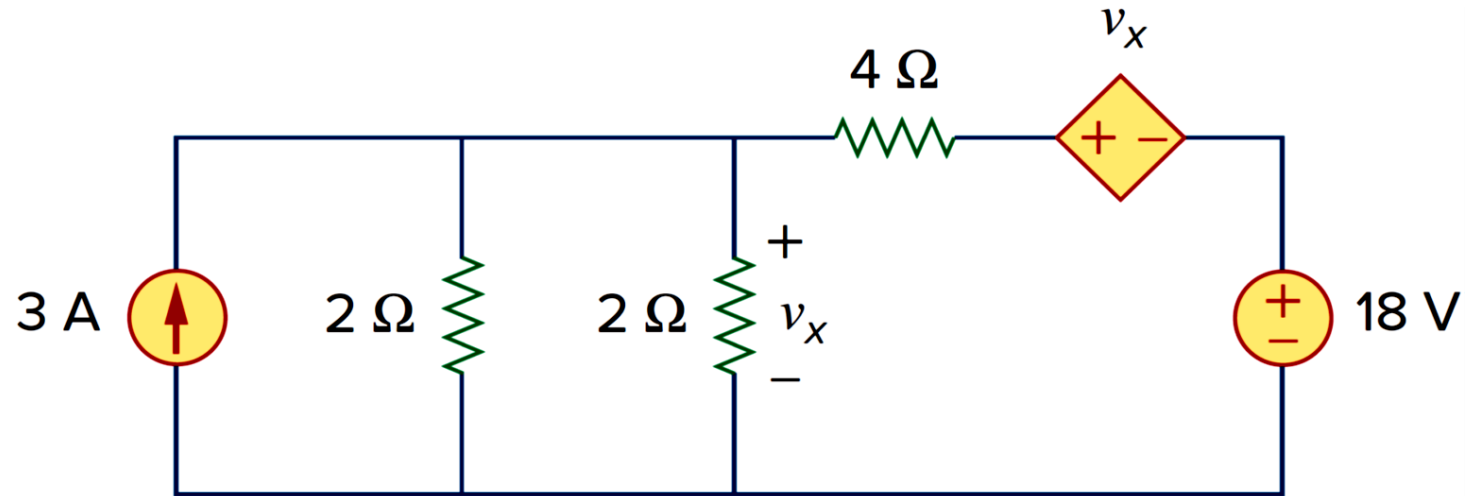
Q7. Find v_x in the given figure using source transformation.



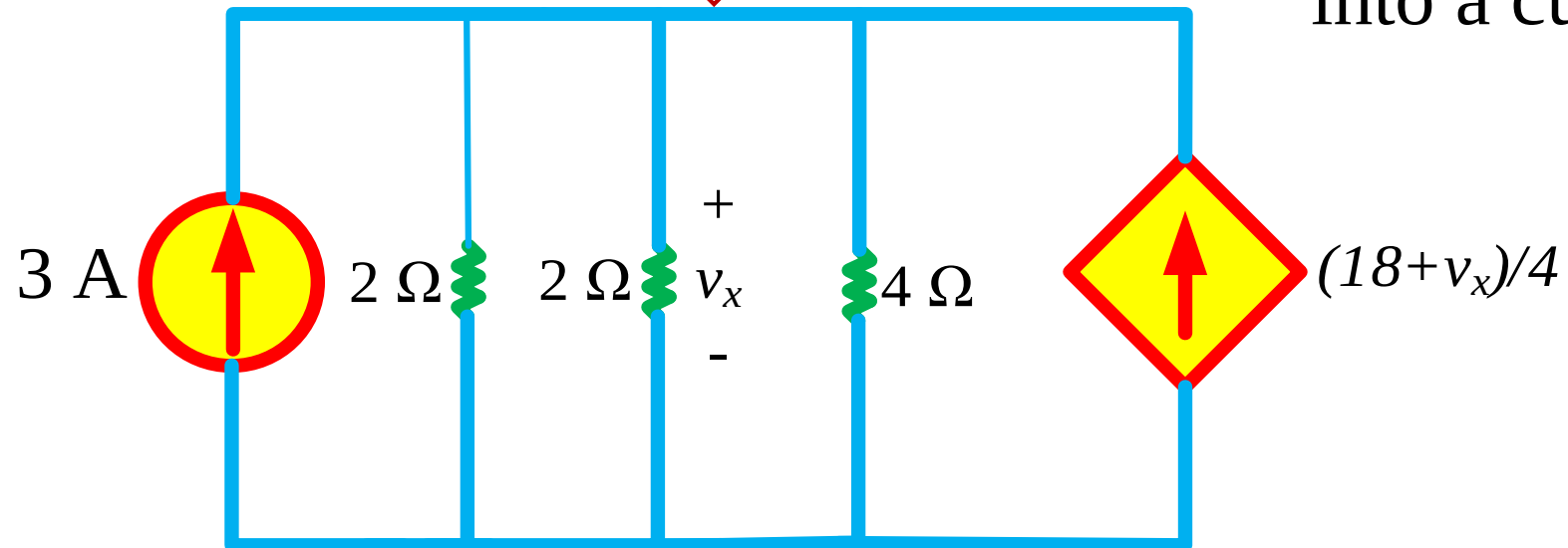
Answer:

Converting VCCS $0.25 v_x$ in parallel with 4Ω into a voltage source will bring 18 V independent voltage source in series.

EXERCISE / NUMERICAL ANALYSIS

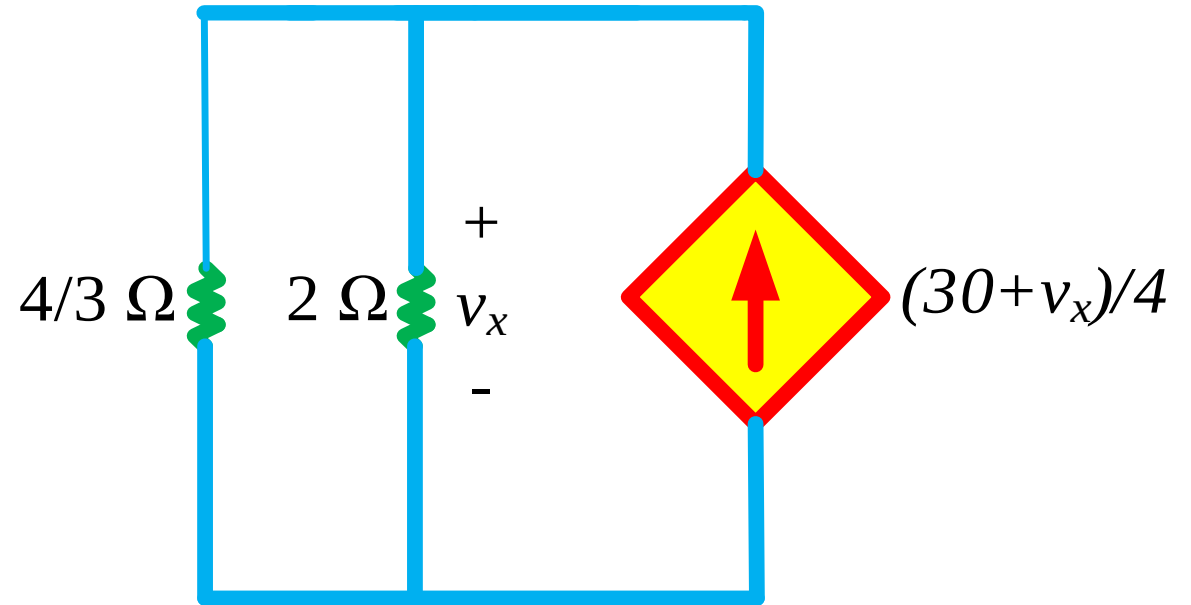


Converting $(18+v_x)$ V voltage source into a current source



EXERCISE / NUMERICAL ANALYSIS

Reducing resistances and
current sources in parallel



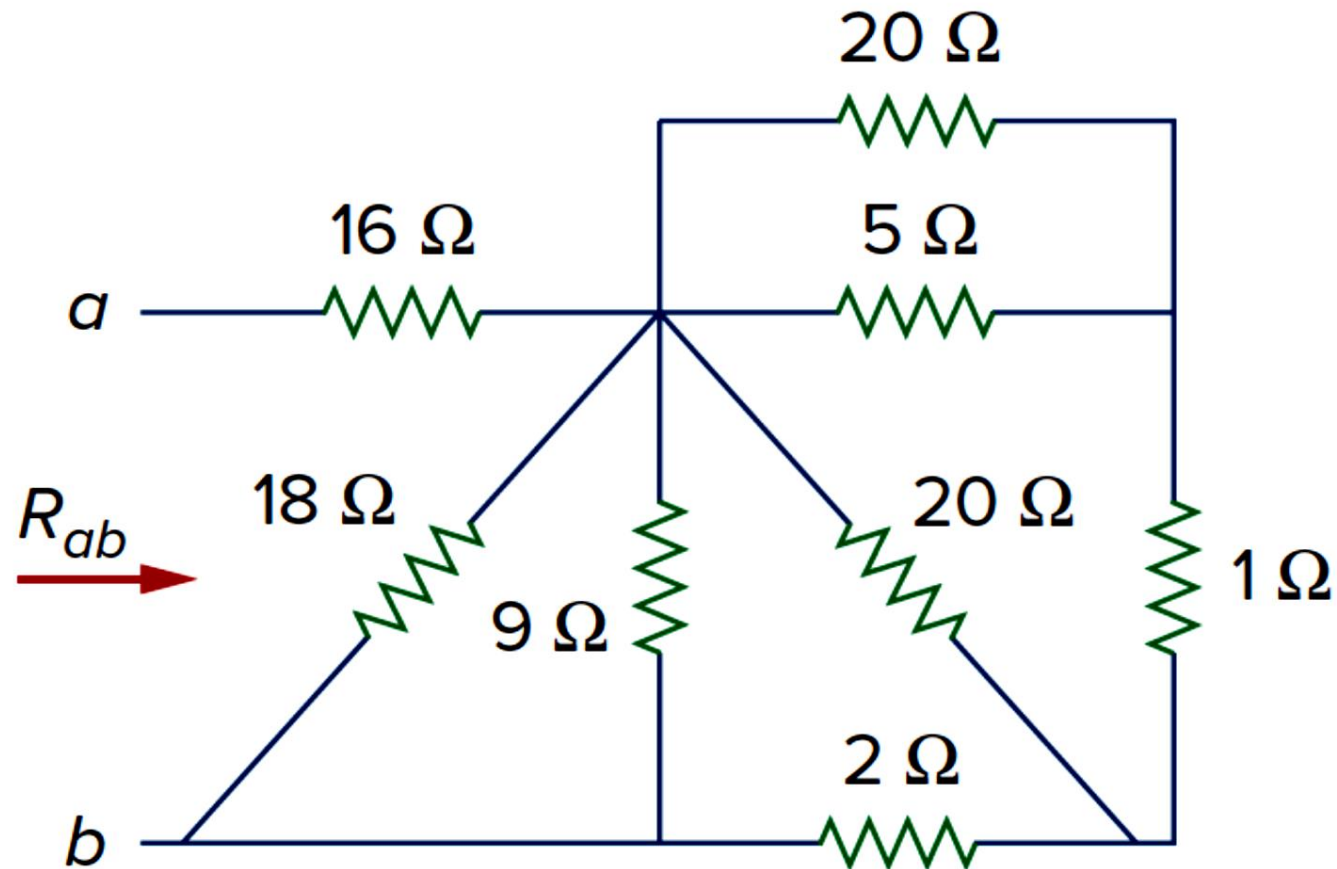
Using Current division,

$$\frac{(30 + v_x)}{4} \times \frac{\left(\frac{4}{3}\right)}{\left(\frac{4}{3}\right) + 2} \times 2 = v_x \Rightarrow 30 + v_x = 5v_x$$

$$\Rightarrow v_x = 7.5V$$

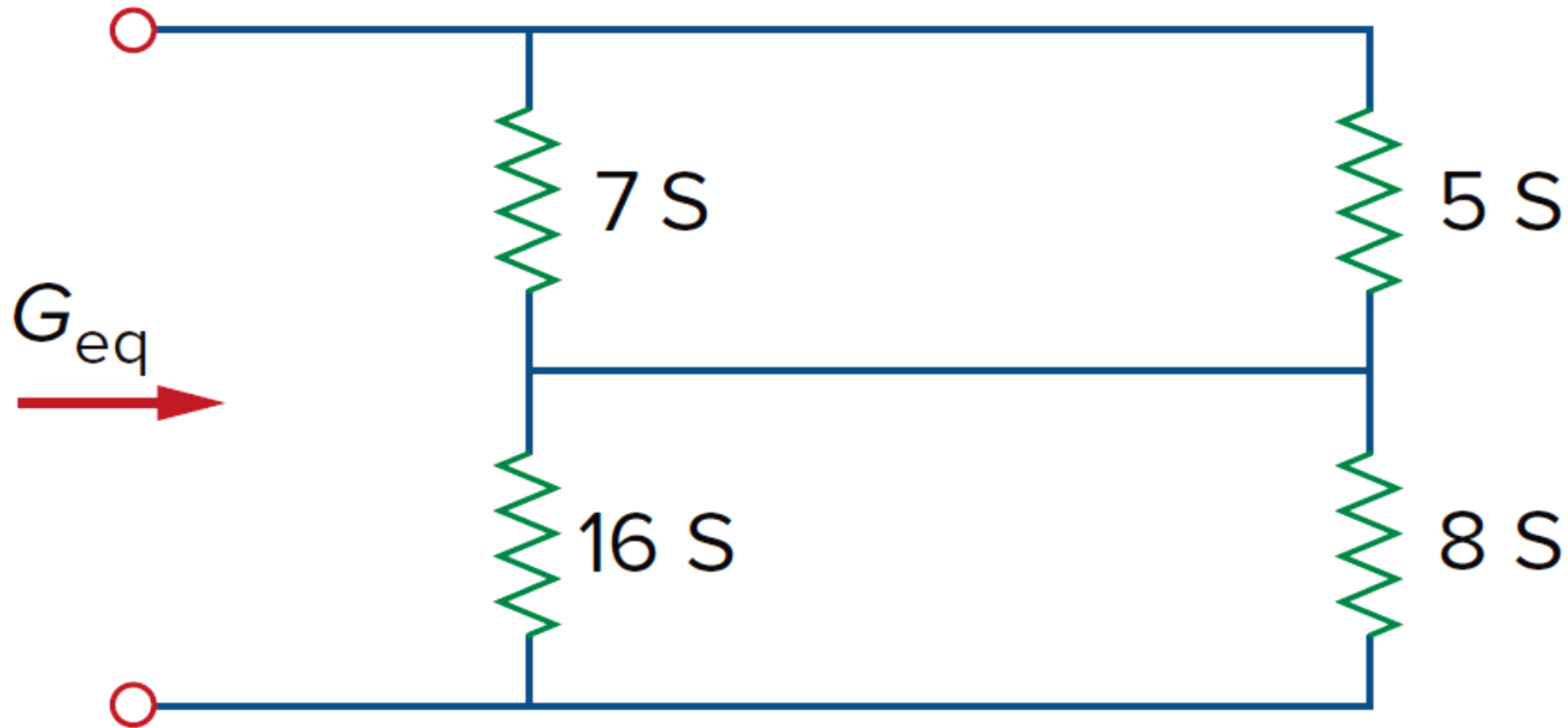
UNSOLVED PROBLEMS

Q1. Find R_{ab} for the circuit given below. [**Answer = 19 Ω**]



UNSOLVED PROBLEMS

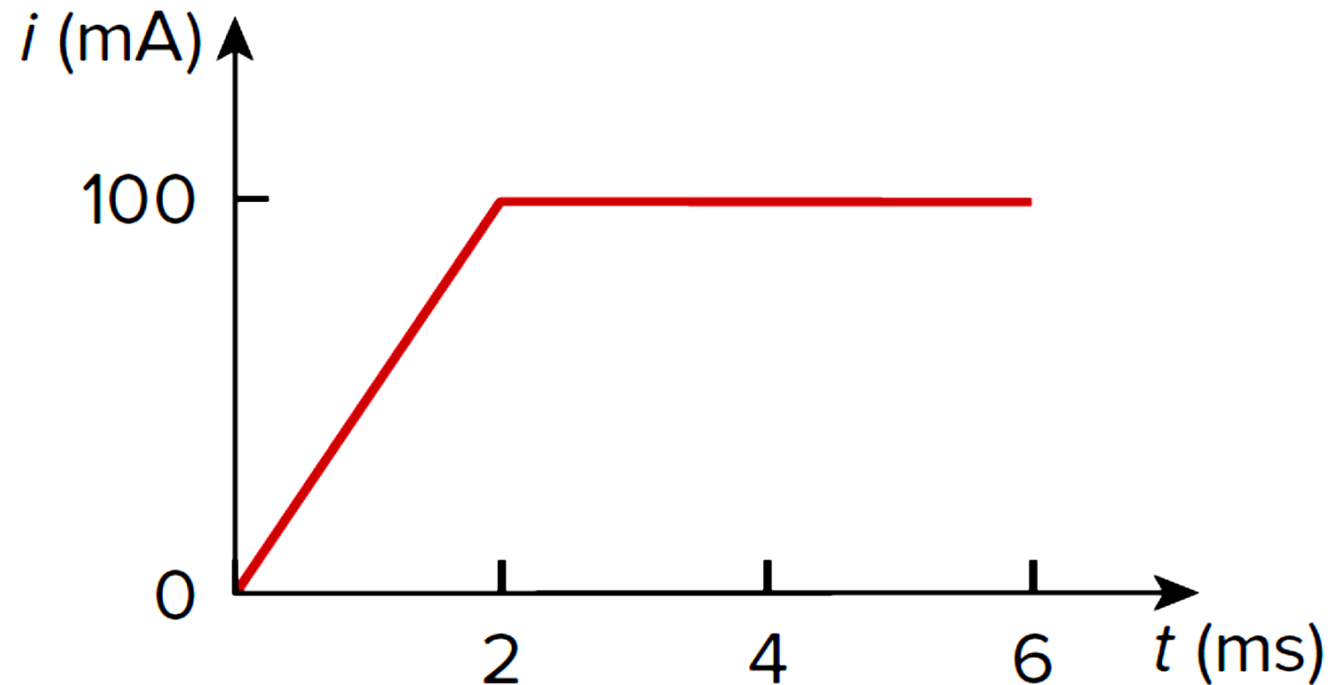
Q2. Calculate G_{eq} in the circuit given below. [**Answer = 8 S**]



UNSOLVED PROBLEMS

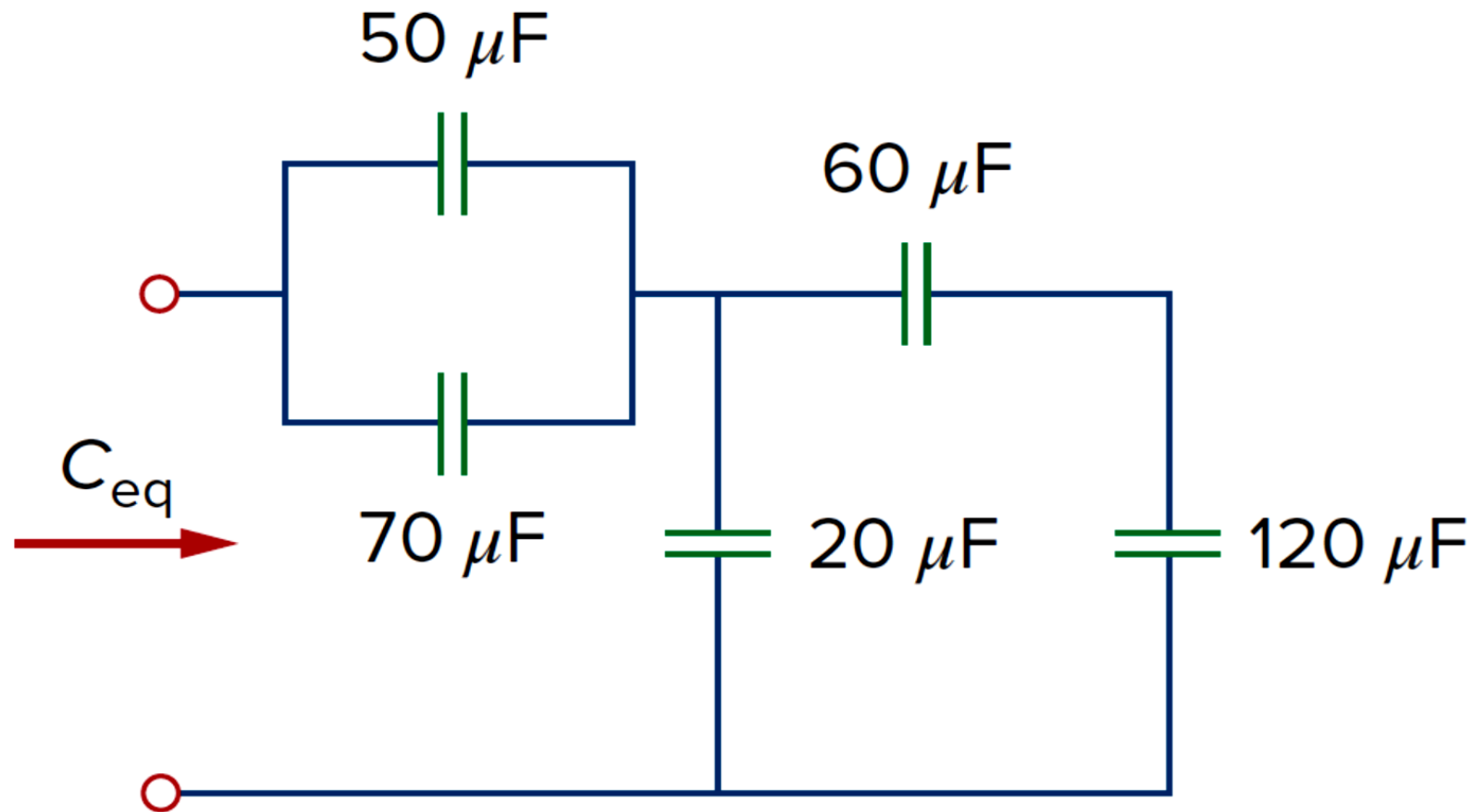
Q3. An initially uncharged 1-mF capacitor has the current as shown in figure below. Calculate the voltage across it at $t = 2$ ms and $t = 5$ ms.

[Answer = 100 mV and 400 mV]



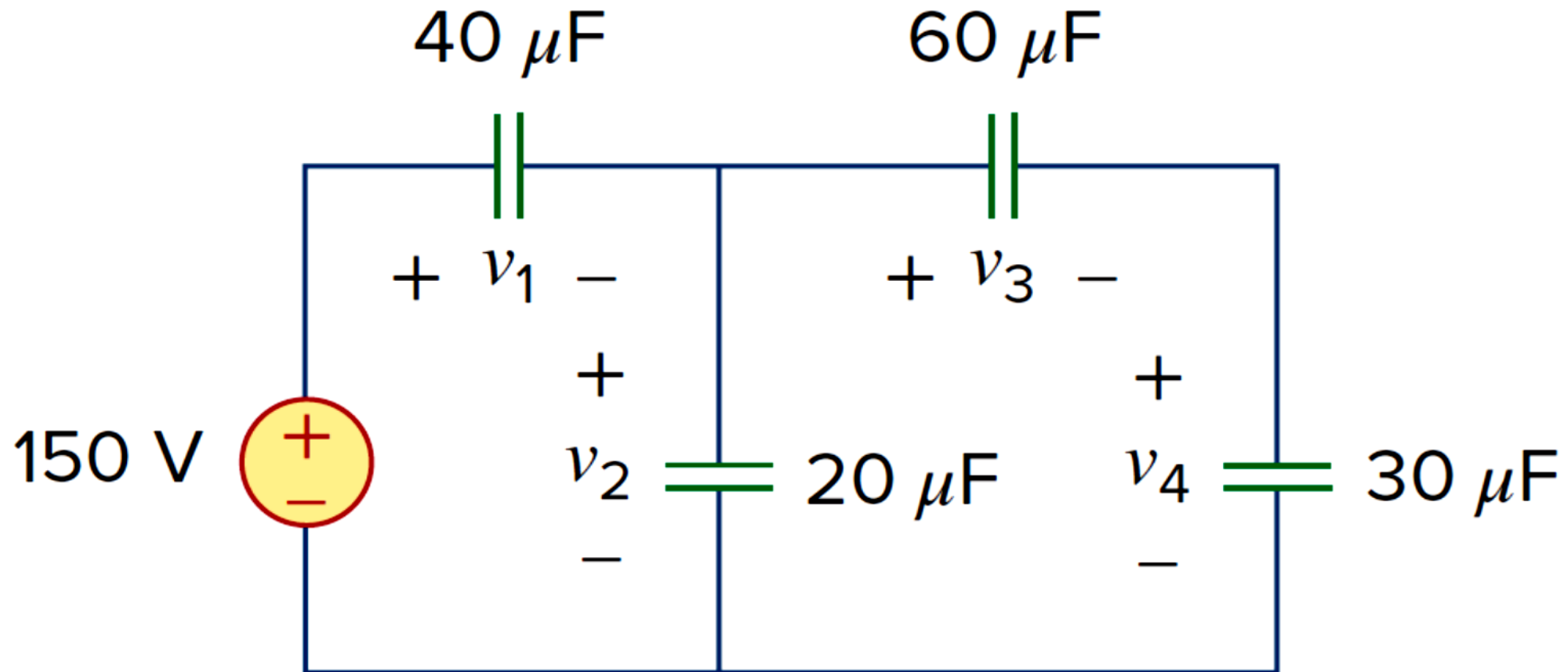
UNSOLVED PROBLEMS

Q4. Find the equivalent capacitance seen at the terminals of the circuit shown below. [Answer = $40\ \mu\text{F}$]



UNSOLVED PROBLEMS

Q5. Find the voltage across each of the capacitors in the circuit shown below. [**Answers:** $v_1 = 75 \text{ V}$, $v_2 = 75 \text{ V}$, $v_3 = 25 \text{ V}$, $v_4 = 50 \text{ V}$]



REFERENCES

1. Alexander, Charles K., and Sadiku, Matthew N. O., “Basic Laws,” in *Fundamentals of Electric Circuits*, 5th Ed, McGraw Hill, Indian Edition, 2013,pp. 29-78.
2. Alexander, Charles K., and Sadiku, Matthew N. O., “Circuit Theorems,” in *Fundamentals of Electric Circuits*, 5th Ed, McGraw Hill, Indian Edition, 2013,pp. 125-171.
3. William H. Hayt , Jack Kemmerly and Steven M. Durbin, “Voltage and Current Laws,” in *Engineering Circuit Analysis*, 8th edition, Published by McGraw-Hill, pp. 39-78